# Transport Properties

## 1. INTRODUCTION

In this chapter we apply the Cartesian-tensor expansion, which was formulated in Chapter VII, to the calculation of transport properties. We shall consider in detail the transport properties which relate the electron current density and heat flux to electric and magnetic fields and to spatial gradients. We shall not consider the contribution of the electrons to the overall viscous stress, since this contribution is almost always small. Our emphasis on the electron transport properties stems from the important role of electrons in plasma transport phenomena and from the particular simplifications which can be used in the calculation of these properties.

Under most conditions, transport phenomena associated with the heavy particles in a plasma can be satisfactorily described by the Chapman-Enskog theory as developed for mixtures of neutral gases, and the corresponding transport properties can be calculated by the use of the classical methods of Hirschfelder, Curtiss, and Bird (1964). Although it is possible to apply these same methods to the evaluation of certain electron transport properties as well, the resultant calculations are more complicated than necessary. Furthermore, for the electrons we wish to consider in addition the effects of magnetic fields and high-frequency electric fields and to develop a rigorous theory that applies when the electron and heavy-particle temperatures differ.

We consider first, in Sec. 2, the electron transport properties of a weakly ionized plasma. In this case, for which electron-electron collisions may be neglected, the calculations are simplest. Qualitatively, however, the resulting expressions for the electron current density and heat flux are in accord with those that apply to a partially ionized plasma. In Sec. 3 we discuss the conditions for which the isotropic part of the electron distribution function is Maxwellian. Then in Sec. 4 we derive expressions for the electron transport properties of partially ionized plasmas. The transport properties of fully ionized plasmas and the heavy-particle transport

properties are discussed briefly in Secs. 5 and 6. In Sec. 7 we consider various mixture rules that are useful for approximate calculations of electron transport properties and assess the accuracy of these formulas in comparison with the results of the rigorous theory of Sec. 4. Finally, in Sec. 8 we discuss the effects of nonelastic collisions.

## 2. WEAKLY IONIZED PLASMAS

When the degree of ionization of a plasma is sufficiently low, the electron-electron collision terms can be neglected in the electron Boltzmann equation and in the Cartesian-tensor equations of Chapter VII, Sec. 6. We refer to this situation as the *Lorentzian limit*. In this section we shall derive expressions for the electron transport properties that apply to such weakly ionized plasmas. As might be expected, omission of the electron-electron collision terms results in a considerable simplification in the calculation of transport properties. Furthermore, it is possible to obtain from a consideration of the Lorentzian limit the forms for the electron current density and heat flux that remain valid for arbitrary degrees of ionization, so that the discussion of this section serves as useful introduction to the general theory of transport properties. In fact, if electron-ion collisions are retained, the Lorentzian expressions give reasonable values for certain transport properties of partially and fully ionized plasmas. Thus, for example, for a fully ionized plasma the Lorentzian value of the electrical conductivity in the absence of a magnetic field differs by only a factor of approximately two from the exact value.

From equations (VII 6.8) it can be seen that the electron diffusion velocity, and hence current density, and the electron heat flux depend on the anisotropic part $\mathbf{f}^1$ in the Cartesian-tensor expansion of the electron distribution function. Since $\mathbf{f}^1$ is governed by the second of the Cartesian tensor equations, the "momentum" equation (VII 6.40), we consider first the solution of that equation, regarding for the present the isotropic part $f^0$ of the distribution function as unspecified.

The conditions under which the Lorentzian limit applies to equation (VII 6.40) can be estimated from equations (VII 6.35) and (VII 6.36) for the collision terms $\mathbb{C}^1_{eh}$ and $\mathbb{C}^1_{ee}$. From these equations it follows that with respect to order of magnitude we have

$$\frac{|\mathbb{C}^1_{ee}|}{|\mathbb{C}^1_{eh}|} \sim \frac{n_e \Gamma_{ee} 8\pi f^0 f^1}{\nu^{(1)}_{eh} f^1 + \nu^{(1)}_{eh} U_h \, \partial f^0 / \partial C}. \tag{2.1}$$

Here, since all terms in equation (VII 6.36) for $\mathbb{C}^1_{ee}$ are of similar magnitude, we have based our estimate of $|\mathbb{C}^1_{ee}|$ on the first term. We require only

a sufficient condition that the ratio (2.1) be small. Accordingly, we can for simplicity omit the second term in the denominator for the purposes of our estimate. The relative importance of this term will be discussed later in this section. Since the transport properties depend on electrons of thermal energies, that is, electrons for which $\frac{1}{2}m_e C^2 \sim \frac{3}{2}kT_e$, it is appropriate to evaluate the ratio (2.1) in some average sense for thermal electrons. From the normalization condition

$$1 = \int_{-\infty}^{\infty} f_e \, dC = 4\pi \int_0^{\infty} f^0 C^2 \, dC,$$

we can write for thermal electrons

$$f^0 \sim \frac{1}{\bar{C}_e^3}.$$

In terms of the electron-electron momentum-transfer collision frequency [see equation (VII 4.14)]

$$\nu_{ee}^{(1)} = \frac{4n_e \Gamma_{ee}}{C^3} \sim \frac{4n_e \Gamma_{ee}}{\bar{C}_e^3}$$

the ratio (2.1) can then be expressed as

$$\frac{|\mathbb{C}_{ee}^1|}{|\mathbb{C}_{eh}^1|} \sim \frac{\nu_{ee}^{(1)}}{\nu_{eh}^{(1)}} \sim \frac{\bar{\nu}_{ee}}{\bar{\nu}_{eh}}.$$

Here, to be more definite, we have introduced the energy-weighted mean collision frequencies defined by the equation following (II 6.32). For $\bar{\nu}_{eh}$, we have to lowest order in $m_e/m_h$

$$\bar{\nu}_{eh} = \int_0^{\infty} \frac{\frac{1}{2}m_e C^2}{\frac{3}{2}kT_e} \nu_{eh}^{(1)}(C) f^0 4\pi C^2 \, dC. \tag{2.2a}$$

With the use of equation (VII 4.10), $\bar{\nu}_{ee}$ is given by

$$\bar{\nu}_{ee} = n_e \frac{8\sqrt{\pi}}{3} \left(\frac{m_e}{kT_e}\right)^{3/2} \left(\frac{e^2}{4\pi\epsilon_0 m_e}\right)^2 \ln \Lambda \tag{2.2b}$$

when $f^0$ is taken to be the Maxwellian distribution (3.3). It follows from the foregoing that the electron-electron collision term $\mathbb{C}_{ee}^1$ can be neglected in equation (VII 6.40) for $\mathbf{f}^1$ if

$$\bar{\nu}_{ee} \ll \sum_h \bar{\nu}_{eh}. \tag{2.3}$$

When the condition (2.3) is satisfied, equation (VII 6.40) becomes

$$\frac{\partial}{\partial t}(n_e \mathbf{f}^1) + C\nabla(n_e f^0) + \mathbf{a}' n_e \frac{\partial f^0}{\partial C} - \frac{e}{m_e}\mathbf{B} \times n_e \mathbf{f}^1 = -n_e \sum_h \left[ \nu_{eh}^{(1)} \left( \mathbf{f}^1 + \mathbf{U}_h \frac{\partial f^0}{\partial C} \right) \right]. \tag{2.4}$$

To calculate the transport properties of a Lorentzian plasma, we must first solve equation (2.4) for $\mathbf{f}^1$ and then substitute the result in equations (VII 6.8) for the electron diffusion velocity and heat flux. [Since the contribution of the electrons to the viscous stress of the plasma as a whole is almost always small, we shall not consider the electron viscosity coefficient, which from equations (VII 6.8) depends on $f_{\alpha\beta}^2$].

We first calculate the electrical conductivity of a uniform, steady-state plasma in the presence of constant electric and magnetic fields. Since for a uniform weakly ionized plasma, the diffusion velocity of the neutrals and the contribution of the ions to the sum on the right-hand side of equation (2.4) are negligible, we can omit the part of the collision term in equation (2.4) that involves $\mathbf{U}_h$. Under these conditions $\mathbf{a}' = -e\mathbf{E}'/m_e$ [see equation (VII 6.15)], and equation (2.4) becomes

$$\frac{e}{m_e}\mathbf{E}' \frac{\partial f^0}{\partial C} + \frac{e}{m_e}\mathbf{B} \times \mathbf{f}^1 = \mathbf{f}^1 \sum_h \nu_{eh}^{(1)}. \tag{2.5}$$

Equation (2.5) can be solved directly for $\mathbf{f}^1$ in terms of $f^0$. For this purpose we adopt the "natural" coordinate system defined by the three orthogonal vectors

$$\mathbf{E}'_{\parallel} \equiv (\mathbf{E} \cdot \mathbf{b})\mathbf{b},$$
$$\mathbf{E}'_{\perp} \equiv \mathbf{E}' - \mathbf{E}'_{\parallel},$$

and

$$\mathbf{b} \times \mathbf{E}' \tag{2.6}$$

Here

$$\mathbf{b} \equiv \mathbf{B}/B \tag{2.7}$$

is a unit vector the direction of the magnetic field. The vector $\mathbf{f}^1$ can then be written in terms of its components in these directions as

$$\mathbf{f}^1 = [f_{\parallel}^1 \mathbf{E}'_{\parallel} + f_{\perp}^1 \mathbf{E}'_{\perp} + f_H^1(\mathbf{b} \times \mathbf{E}')](e/m_e), \tag{2.8}$$

where the factor $e/m_e$ is introduced for later convenience. The subscripts $\parallel$, $\perp$, and $H$ denote the directions parallel to and perpendicular to the

magnetic field, and the "Hall" direction, which is perpendicular to both $\mathbf{B}$ and $\mathbf{E}'$. The coefficients $f_{\parallel}^1, f_{\perp}^1, f_H^1$ can be found from the substitution of $\mathbf{f}^1$ in the form (2.8) into equation (2.5). This yields, after division by $e/m_e$,

$$(\mathbf{E}'_{\parallel} + \mathbf{E}'_{\perp})\frac{\partial f^0}{\partial C} + (\omega_e f_{\perp}^1 \mathbf{b} \times \mathbf{E}' - \omega_e f_H^1 \mathbf{E}'_{\perp}) = \nu_{eH}(f_{\parallel}^1 \mathbf{E}'_{\parallel} + f_{\perp}^1 \mathbf{E}'_{\perp} + f_H^1 \mathbf{b} \times \mathbf{E}')$$

where the electron cyclotron frequency $\omega_e$ is

$$\omega_e = eB/m_e \tag{2.9}$$

and where

$$\nu_{eH} \equiv \sum_h \nu_{eh}^{(1)}. \tag{2.10}$$

Equating coefficients of the three orthogonal vectors, we obtain

$$f_{\parallel}^1 = \frac{1}{\nu_{eH}}\frac{\partial f^0}{\partial C},$$

$$\nu_{eH} f_{\perp}^1 + \omega_e f_H^1 = \frac{\partial f^0}{\partial C}, \tag{2.11a}$$

and

$$\nu_{eH} f_H^1 = \omega_e f_{\perp}^1.$$

From the last two of these relations, $f_{\perp}^1$ and $f_H^1$ are given by

$$f_{\perp}^1 = \frac{\nu_{eH}}{\nu_{eH}^2 + \omega_e^2}\frac{\partial f^0}{\partial C} \tag{2.11b}$$

and

$$f_H^1 = \frac{\omega_e}{\nu_{eH}^2 + \omega_e^2}\frac{\partial f^0}{\partial C}. \tag{2.11c}$$

With these results and equation (VII 6.8) for the electron diffusion velocity $\mathbf{U}_e$, we can now calculate the contribution $\mathbf{J}_e$ of the electrons to the current-density vector $\mathbf{J}$. Thus we have

$$\mathbf{J}_e \equiv -en_e\mathbf{U}_e = -en_e\frac{4\pi}{3}\int_0^\infty C^3\mathbf{f}^1\,dC. \tag{2.12}$$

As indicated in Chapter IV, Sec. 3, except for the case of ion slip in very strong magnetic fields, the ion current density for a uniform plasma is much less than the electron current density, so that $\mathbf{J} \simeq \mathbf{J}_e$. (The ion

current density can also be important in the region of nonuniform plasma in the vicinity of a solid surface; see Chapter III, Secs. 6 and 7.) Over a wide range of conditions, then, to calculate the overall electrical conductivity of a weakly ionized plasma we need consider only the foregoing solution of the electron Boltzmann equation, which is mathematically decoupled from the Boltzmann equations for the other species. The result from the substitution of equations (2.8) and (2.11) in equation (2.12) for $\mathbf{J}_e$ can be expressed in terms of the following Ohm's-law relation [cf. equation (IV 3.21a)]:

$$\mathbf{J}_e = \sigma_{\parallel} \mathbf{E}'_{\parallel} + \sigma_{\perp} \mathbf{E}'_{\perp} + \sigma_H \mathbf{b} \times \mathbf{E}'. \tag{2.13}$$

The components $\sigma_{\parallel}$, $\sigma_{\perp}$, and $\sigma_H$ of the so-called tensor conductivity are then

$$\sigma_{\parallel} = \frac{4\pi}{3} \frac{n_e e^2}{m_e} \int_0^{\infty} \frac{C^3}{\nu_{eH}} \left( -\frac{\partial f^0}{\partial C} \right) dC, \tag{2.14a}$$

$$\sigma_{\perp} = \frac{4\pi}{3} \frac{n_e e^2}{m_e} \int_0^{\infty} \frac{C^3 \nu_{eH}}{\nu_{eH}^2 + \omega_e^2} \left( -\frac{\partial f^0}{\partial C} \right) dC, \tag{2.14b}$$

and

$$\sigma_H = \frac{4\pi}{3} \frac{n_e e^2}{m_e} \int_0^{\infty} \frac{C^3 \omega_e}{\nu_{eH}^2 + \omega_e^2} \left( -\frac{\partial f^0}{\partial C} \right) dC. \tag{2.14c}$$

In the absence of a magnetic field, $\omega_e = 0$ and the perpendicular and parallel conductivities $\sigma_{\perp}$ and $\sigma_{\parallel}$ are equal. In this case the current density is given by

$$\mathbf{J}_e = \sigma_{\parallel} \mathbf{E}.$$

Thus, the conventional scalar conductivity is identical to $\sigma_{\parallel}$ and, in the Lorentzian limit, can be found from equation (2.14a).

The evaluation of expressions (2.14) for the Lorentzian conductivity requires a knowledge of the isotropic part $f^0$ of the distribution function. Since from equations (2.11) $\mathbf{f}^1$ is now known in terms of $f^0$, the function $f^0$ can be obtained from the first of the Cartesian-tensor equations, equation (VII 6.39). Under the present conditions for a weakly ionized plasma, this equation becomes

$$-\frac{e}{3m_e} \mathbf{E}' \cdot \frac{d}{dC} (C^2 \mathbf{f}^1) = \sum_h \frac{m_e}{m_h} \frac{d}{dC} \left[ C^3 \left( f^0 + \frac{kT_h}{m_e} \frac{1}{C} \frac{df^0}{dC} \right) \nu_{eh}^{(1)} \right]. \tag{2.15}$$

Here we have omitted electron-electron collision term $\mathbb{C}_e^1$ in comparison with the electron/heavy-particle collision terms $\mathbb{C}_{eh}^1$. By means of an estimate similar to that used to obtain the relative magnitudes (2.2) of $\mathbb{C}_{ee}^1$ and

$\mathbb{C}^1_{eh}$ [see equation (3.6) in the following section], the condition that $\mathbb{C}^0_{ee}$ be neglected in equation (2.15) is

$$\bar{\nu}_{ee} \ll \sum_h \frac{m_e}{m_h} \bar{\nu}_{eh}. \tag{2.16}$$

Clearly this condition is much more restrictive than the condition (2.3) for the validity of the foregoing Lorentzian analysis for $\mathbf{f}^1$.

To solve equation (2.15) for $f^0$ we must first evaluate $\mathbf{E}' \cdot \mathbf{f}^1$. From equations (2.8) and (2.11) we have

$$\begin{aligned}\mathbf{E}' \cdot \mathbf{f}^1 &= \frac{e}{m_e}(f^1_{\parallel} E'_{\parallel} + f^1_{\perp} E'_{\perp}) \\ &= \frac{e}{m_e}\left(E'^2_{\parallel} + \frac{\nu^2_{eH}}{\nu^2_{eH} + \omega^2_e} E'^2_{\perp}\right) \frac{1}{\nu_{eH}} \frac{df^0}{dC}.\end{aligned}$$

With the use of this result, integration of equation (2.15) with respect to $C$ gives

$$\frac{df^0}{dC} = \frac{-\sum_h (m_e/m_h)\nu^{(1)}_{eh} C f^0}{\sum_h (kT_h/m_h)\nu^{(1)}_{eh} + \frac{1}{3}(e^2/m_e^2)[E'^2_{\parallel} + E'^2_{\perp}\nu^2_{eH}/(\nu^2_{eH} + \omega^2_e)]1/\nu_{eH}}, \tag{2.17}$$

where the constant of integration has been set equal to zero on the basis of the condition that both $f^0$ and $df^0/dC$ vanish as $C \to \infty$. Equation (2.17) can be formally integrated to yield, for a value of $T_h$ that is the same for all heavy species,

$$f^0(C) = f^0(0)\exp\left\{-\int_0^C \frac{m_e C\, dC}{kT_h + \frac{1}{3}(e^2/m_e^2\nu_{eH})[E'^2_{\parallel} + E'^2_{\perp}\nu^2_{eH}/(\nu^2_{eH} + \omega^2_e)]1/\sum_h (\nu^{(1)}_{eh}/m_h)}\right\}. \tag{2.18}$$

The integration constant $f^0(0)$ can be found from the normalization condition

$$\int_{-\infty}^{\infty} f_e(\mathbf{C})\, d^3C = 4\pi \int_0^{\infty} f^0(C) C^2\, dC = 1.$$

[From the first of equations (VII 6.19), terms other than $f^0$ in the Cartesian-tensor expansion do not contribute to this integral.]

In equation (2.18) for $f^0$, when the field $\mathbf{E}'$ is sufficiently small the second term in the denominator of the integral can be neglected. In this case $f^0$ reduces to a Maxwellian function with the electron temperature equal to the heavy-particle temperature. For strong fields, the electron temperature is greater than $T_h$ and $f^0$ is not, in general, Maxwellian.

To evaluate explicitly the Lorentzian distribution (2.18) and the conductivity expressions (2.14), particular forms must be taken for $\nu_{eh}^{(1)}$ (see, for example, Chapman and Cowling, 1952, Secs. 18.71 and 18.73).

We now return to equation (2.4) for $\mathbf{f}^1$ and consider separately the effects on the solution of that equation of heavy-particle diffusion velocities, high-frequency electric fields, and spatial gradients.

For a uniform, steady-state plasma retention of the part of the collision term in equation (2.4) that involves the heavy-particle diffusion velocities $\mathbf{U}_h$ results in the addition of a term

$$\sum_h \nu_{eh}^{(1)} \mathbf{U}_h \frac{\partial f^0}{\partial C} \tag{2.19}$$

to equation (2.5). Since this equation is linear in $\mathbf{f}^1$, the solution for $\mathbf{f}^1$ can be written as the sum of the previous solution [(2.8) and (2.11)], which depends on the electric field, and the solution of the equation

$$-\sum_h \nu_{eh}^{(1)} \mathbf{U}_h \frac{\partial f^0}{\partial C} + \frac{e}{m_e} \mathbf{B} \times \mathbf{f}^1 = \mathbf{f}^1 \sum_h \nu_{eh}^{(1)}.$$

By means of an analysis similar to that used to obtain equations (2.8) and (2.11), this solution is

$$\mathbf{f}^1 = -\sum_h \nu_{eh}^{(1)} [f_\parallel^1 (\mathbf{U}_h)_\parallel + f_\perp^1 (\mathbf{U}_h)_\perp + f_H^1 \mathbf{b} \times \mathbf{U}_h], \tag{2.20}$$

where

$$(\mathbf{U}_h)_\parallel \equiv (\mathbf{U}_h \cdot \mathbf{b}) b$$

and

$$(\mathbf{U}_h)_\perp \equiv \mathbf{U}_h - (\mathbf{U}_h)_\parallel,$$

and where $f_\parallel^1$, $f_\perp^1$, and $f_H^1$ are again given by equations (2.11). From this result and equation (2.12) for the electron diffusion velocity $\mathbf{U}_e$, we can estimate the contribution to $\mathbf{U}_e$ that results from the retention of $\mathbf{U}_h$ in the collision term in equation (2.4). For a particular heavy-particle species this contribution is of the order of (see Exercise 2.2)

$$\begin{aligned} (\mathbf{U}_e)_\parallel &\sim \frac{\nu_{eh}^{(1)}}{\nu_{eH}} (\mathbf{U}_h)_\parallel, \\ (\mathbf{U}_e)_\perp &\sim \frac{1}{1+\beta_e^2} \frac{\nu_{eh}^{(1)}}{\nu_{eH}} (U_h)_\perp, \end{aligned} \tag{2.21}$$

and

$$(\mathbf{U}_e)_H \sim \frac{\beta_e}{1+\beta_e^2} \frac{\nu_{eh}^{(1)}}{\nu_{eH}} \mathbf{b} \times \mathbf{U}_h,$$

where

$$\beta_e \sim \frac{\omega_e}{\nu_{eH}^{(1)}}.$$

To the extent that the heavy-particle diffusion velocities are small, this effect can then be neglected. Accordingly, we shall omit the parts of $\mathbb{C}_{eh}^1$ that involves $\mathbf{U}_h$ in our further calculations.

We now consider the technically more important problem of the response of a uniform, weakly ionized plasma to a high-frequency electric field (see Chapter III, Sec. 8). In this analysis, we include the effect of a constant magnetic field. If it is assumed that high-frequency time variation of the electron number density can be neglected, the pertinent form of equation (2.4) for $\mathbf{f}^1$ is

$$\frac{\partial \mathbf{f}^1}{\partial t} - \frac{e}{m_e}\mathbf{E}'\frac{\partial f^0}{\partial C} - \frac{e}{m_e}\mathbf{B} \times \mathbf{f}^1 = -\mathbf{f}^1 \sum_h \nu_{eh}^{(1)}. \tag{2.22}$$

We now represent the electric field by

$$\mathbf{E}' = \mathbf{E}^0 + \mathbf{E}^1 \operatorname{Re} e^{-i\omega t},$$

where $i^2 = -1$, $\mathbf{E}^0$ is a constant (d.c) field, and $\mathbf{E}^1$ is the (real) amplitude of the component of the field of frequency $\omega$. In accord with our previous solution (2.18) for $f^0$, if $\mathbf{E}^1$ is sufficiently small we can neglect the time variation of $f^0$ in equation (2.22). Our previous solution (2.8) and (2.11) then applies to the part of $\mathbf{f}^1$ associated with the constant field $\mathbf{E}^0$, and the corresponding current density is given by equations (2.13) and (2.14), as before. The additional effect of the high-frequency field is obtained from the solution of equation (2.22) with $\mathbf{E}'$ replaced by

$$\mathbf{E}' = \mathbf{E}^1 \operatorname{Re} e^{-i\omega t}. \tag{2.23}$$

To effect this solution we write for the high-frequency part of $\mathbf{f}^1$

$$\mathbf{f}^1 = \operatorname{Re}\left\{[\tilde{f}_{\parallel}^1 \mathbf{E}_{\parallel}^1 + \tilde{f}_{\perp}^1 \mathbf{E}_{\perp}^1 + \tilde{f}_H^1 \mathbf{b} \times \mathbf{E}^1]\frac{e}{m_e} e^{-i\omega t}\right\}. \tag{2.24}$$

where $\mathbf{E}_{\parallel}^1 \equiv (\mathbf{E}^1 \cdot \mathbf{b})\mathbf{b}$ and $\mathbf{E}_{\perp}^1 \equiv \mathbf{E}^1 - \mathbf{E}_{\parallel}^1$. The complex coefficients $\tilde{f}_{\parallel}^1$, $\tilde{f}_{\perp}^1$, and $\tilde{f}_H^1$ can be found from the substitution of equations (2.23) and (2.24) in equation (2.22). Equating coefficients in the directions of the three orthogonal vectors $\mathbf{E}_{\parallel}^1$, $\mathbf{E}_{\perp}^1$, and $\mathbf{b} \times \mathbf{E}^1$, we thus obtain

$$(-i\omega + \nu_{eH})\tilde{f}_{\parallel}^1 = \frac{\partial f^0}{\partial C},$$

$$(-i\omega + \nu_{eH})\tilde{f}_{\perp}^1 + \omega_e \tilde{f}_H^1 = \frac{\partial f^0}{\partial C},$$

and

$$(-i\omega + \nu_{eH})\tilde{f}_H^1 = \omega_e \tilde{f}_\perp^1 .$$

Again we have used the total collision frequency $\nu_{eH} \equiv \sum_h \nu_{eh}^{(1)}$. The solution of preceding equations for the three components of $f^1$ can be expressed as follows:

$$\tilde{f}_\parallel^1 = \frac{\nu_{eH} + i\omega}{\nu_{eH}^2 + \omega^2} \frac{\partial f^0}{\partial C},$$

$$\begin{aligned} \tilde{f}_H^1 &= \frac{\omega_e}{(\nu_{eH} - i\omega)^2 + \omega_e^2} \frac{\partial f^0}{\partial C} \\ &= \frac{-i}{2}\left[\frac{1}{\nu_{eH} - i(\omega + \omega_e)} - \frac{1}{\nu_{eH} - i(\omega - \omega_e)}\right]\frac{\partial f^0}{\partial C} \\ &= \frac{1}{2}\left[\frac{(\omega + \omega_e) - i\nu_{eH}}{\nu_{eH}^2 + (\omega + \omega_e)^2} - \frac{(\omega - \omega_e) - i\nu_{eH}}{\nu_{eH}^2 + (\omega - \omega_e)^2}\right]\frac{\partial f^0}{\partial C}. \end{aligned} \tag{2.25}$$

and

$$\begin{aligned} \tilde{f}_\perp^1 &= \frac{\nu_{eH} - i\omega}{\omega_e} \tilde{f}_H^1 \\ &= \frac{1}{2}\left[\frac{\nu_{eH} + i(\omega + \omega_e)}{\nu_{eH}^2 + (\omega + \omega_e)^2} + \frac{\nu_{eH} + i(\omega - \omega_e)}{\nu_{eH}^2 + (\omega - \omega_e)^2}\right]\frac{\partial f^0}{\partial C}. \end{aligned}$$

Here the final forms for $\tilde{f}_H^1$ and $\tilde{f}_\perp^1$ have been written with our previous equations (2.11) in mind, for reasons that will become apparent shortly.

With the use of equation (2.12) for $\mathbf{J}_e$, the current density associated with the high-frequency field can be expressed in terms of complex conductivities $\tilde{\sigma}_\parallel$, $\tilde{\sigma}_\perp$, and $\tilde{\sigma}_H$ as

$$\mathbf{J}_e = \mathrm{Re}[(\tilde{\sigma}_\parallel \mathbf{E}_\parallel^1 + \tilde{\sigma}_\perp \mathbf{E}_\perp^1 + \tilde{\sigma}_H \mathbf{b} \times \mathbf{E}^1)e^{-i\omega t}]. \tag{2.26}$$

Thus, for example, the real and imaginary parts of $\tilde{\sigma}_\parallel$ determine respectively the in-phase and out-phase components of $\mathbf{J}_e^\parallel$. In general, the total current density is the sum of the current densities given by equations (2.26) and (2.13).

Comparison of equations (2.25) and (2.11) shows that the present complex conductivities can be written in terms of our previous dc conductivities (2.14) in the forms

$$\begin{aligned} \tilde{\sigma}_\parallel &= \sigma_\perp(\omega) + i\sigma_H(\omega), \\ \tilde{\sigma}_\perp &= \tfrac{1}{2}\{\sigma_\perp(\omega + \omega_e) + \sigma_\perp(\omega - \omega_e) + i[\sigma_H(\omega + \omega_e) + \sigma_H(\omega - \omega_e)]\}, \end{aligned} \tag{2.27}$$

and

$$\tilde{\sigma}_H = \tfrac{1}{2}\{\sigma_H(\omega + \omega_e) - \sigma_H(\omega - \omega_e) - i[\sigma_\perp(\omega + \omega_e) - \sigma_\perp(\omega - \omega_e)]\}.$$

Here, for example, $\sigma_\perp(\omega + \omega_e)$ represents the value of $\sigma_\perp$ as given by equation (2.14b) with $\omega_e$ in that equation replaced by $\omega + \omega_e$. Similar interpretations apply to the other terms in equations (2.27). It follows that to obtain the current density as a result of both constant and high-frequency electric fields and in the presence of a constant magnetic field, it is necessary only to calculate $\sigma_\perp$ and $\sigma_H$ as functions of $\omega_e$ from equations (2.14b) and (2.14c). (As we have already seen, $\sigma_\parallel$ is equal to the value of $\sigma_\perp$ when $\omega_e = 0$). It turns out, in fact, that for partially and fully ionized plasmas, where the effects of electron-electron collisions must be included, the present result in the form of equations (2.27) still applies, although the values of $\sigma_\perp$ and $\sigma_H$ are modified from those given by equations (2.14). With regard to the propagation of electromagnetic waves, equation (2.2b) in the $\mathbf{B} = 0$ limit

$$\mathbf{J}_e = \mathrm{Re}\, \tilde{\sigma}_\parallel \mathbf{E}^1_\parallel e^{-i\omega t}$$

can be used in place of equation (III 8.4b) to obtain a more accurate form of the dispersion relation (III 8.5a).

We now consider the effects of spatial gradients. As before, we analyze these effects separately, so that the appropriate form of equation (2.4) for $\mathbf{f}^1$ is

$$-C\nabla(n_e f^0) + \frac{e}{m_e}\mathbf{B} \times n_e \mathbf{f}^1 = n_e \nu_{eH} \mathbf{f}^1. \tag{2.28}$$

By analogy with our previous results, as given for example by equations (2.8) and (2.11), the solution to equation (2.28) is

$$\mathbf{f}^1 = f^1_\parallel \nabla_\parallel \ln (n_e f^0) + f^1_\perp \nabla_\perp \ln (n_e f^0) + f^1_H \mathbf{b} \times \nabla \ln (n_e f^0), \tag{2.29}$$

where

$$f^1_\parallel = -\frac{1}{\nu_{eH}} C f^0,$$

$$f^1_\perp = -\frac{\nu_{eH}}{\nu^2_{eH} + \omega^2_e} C f^0, \tag{2.30}$$

and

$$f^1_H = -\frac{\omega_e}{\nu^2_{eH} + \omega^2_e} C f^0.$$

In equation (2.28) the as-yet-unspecified vector

$$\nabla \ln (n_e f^0) = \frac{\partial}{\partial x_\alpha} \ln (n_e f^0)$$

is regarded as the "driving force" which is analogous to the electric field in our previous calculations. The components of this vector that are parallel to and perpendicular to the magnetic field are denoted in equation (2.29) by $\nabla_\parallel$ and $\nabla_\perp$, respectively.

From the solution (2.29) and (2.30) we can now calculate the electron heat flux which results from the spatial gradient $\nabla \ln (n_e f^0)$. The heat flux vector is given in terms of $\mathbf{f}^1$ by equations (VII 6.8) as

$$\mathbf{q}_e = \frac{2\pi}{3} n_e m_e \int_0^\infty C^5 \mathbf{f}^1 \, dC \tag{2.31}$$

Although it would not be difficult to express $\mathbf{q}_e$ in terms of $\nabla \ln (n_e f^0)$, we shall consider only the particular case where $f^0$ is a Maxwellian function of the electron temperature $T_e$. By direct differentiation of $f_M$ from equation (VII 2.2) we can then write

$$\begin{aligned} \nabla \ln(n_e f^0) &= \nabla \ln(n_e f_M) \\ &= \nabla \ln p_e - \nabla \ln T_e + \nabla \ln f_M \\ &= \nabla \ln p_e + \left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right) \nabla \ln T_e. \end{aligned} \tag{2.32}$$

Here we have introduced the electron pressure $p_e = n_e k T_e$ so that our results will be expressed in a conventional form.

We now substitute $\mathbf{f}^1$ from equations (2.29), (2.30), and (2.32) in equation (2.31) to obtain an expression for $\mathbf{q}_e$. For the moment, we shall consider only the effect of temperature gradients and omit the term $\nabla \ln p_e$ in equation (2.32). The part in question of electron heat flux can then be written in the form

$$\mathbf{q}_e = -\hat{\lambda}_\parallel \nabla_\parallel T_e - \hat{\lambda}_\perp \nabla_\perp T_e - \hat{\lambda}_H \mathbf{b} \times \nabla T_e, \tag{2.33}$$

where the electron thermal-conductivity coefficients are given by

$$\begin{aligned} \hat{\lambda}_\parallel &= \frac{2\pi}{3} \frac{n_e m_e}{T_e} \int_0^\infty \frac{1}{\nu_{eH}} \left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right) C^6 f_M \, dC, \\ \hat{\lambda}_\perp &= \frac{2\pi}{3} \frac{n_e m_e}{T_e} \int_0^\infty \frac{\nu_{eH}}{\nu_{eH}^2 + \omega_e^2} \left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right) C^6 f_M \, dC, \end{aligned} \tag{2.34}$$

and

$$\hat{\lambda}_H = \frac{2\pi}{3}\frac{n_e m_e}{T_e}\int_0^\infty \frac{\omega_e}{\nu_{eH}^2 + \omega_e^2}\left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right)C^6 f_M \, dC.$$

From these results it can be seen that, as in the case of electrical conductivity, a magnetic field acts to reduce the perpendicular thermal conductivity.

In the foregoing we have considered separately the diffusion, and hence current flow, caused by an electric field and the heat flux caused by a temperature gradient. Since equation (2.4) is linear in $\mathbf{f}^1$, we have noted that the consequences of the various driving forces which we have considered are additive. Beyond this, since by equations (2.12) and (2.31) both $\mathbf{J}_e$ and $\mathbf{q}_e$ depend on the same function $\mathbf{f}^1$, there are additional effects of these driving forces. That is, by equation (2.31), the part (2.8) of $\mathbf{f}^1$ that results from an electric field will contribute also to the heat flux and, by equation (2.12), the part (2.29) of $\mathbf{f}^1$ that results from a temperature gradient will contribute also to diffusion. This phenomenon, which is difficult to understand on the basis of simple mean-free-path arguments, is known as thermal diffusion.

It is a straightforward matter to generalize our previous analysis to account for the simultaneous action of fields and spatial gradients and to account for thermal diffusion. In these calculations we shall assume that $f^0$ is a Maxwellian function (VII 2.2) of $T_e$, so that in equations (2.11) and (2.14) we have, by differentiation, $-\partial f^0/\partial C = m_e C f_M/kT_e$. This condition makes our present results more directly useful for our discussion in Sec. 4 of partially ionized plasmas.

If we do not include here the effect of high-frequency electric fields [and the effect of the part of the collision term in equation (2.4) that involves $\mathbf{U}_h$] the appropriate solution to equation (2.4) can be written in the form (see Exercise 2.3)

$$\mathbf{f}^1 = Cf_M[A_1 \nabla_\parallel \ln T_e + A_2 \nabla_\perp \ln T_e + A_3 \mathbf{b} \times \nabla \ln T_e$$
$$- \frac{e}{kT_e}(D_1 \mathscr{E}_\parallel + D_2 \mathscr{E}_\perp + D_3 \mathbf{b} \times \mathscr{E})]. \tag{2.35}$$

Here the "generalized electric field"

$$\mathscr{E} \equiv \mathbf{E}' + \frac{\nabla p_e}{en_e} + \frac{m_e}{e}\frac{D\mathbf{u}}{Dt}$$
$$\simeq \mathbf{E}' + \frac{\nabla p_e}{en_e} \tag{2.36}$$

incorporates the term $\nabla p_e$ from equation (2.32) and the term $D\mathbf{u}/Dt$ in the acceleration $\mathbf{a}'$ in equation (2.4) [see equation (VII 6.15)]. The indicated approximation results from the fact that in almost all situations the last term in $\mathscr{E}$ is negligible.

The coefficients $A_1, A_2, A_3$, and $D_1, D_2, D_3$ in equation (2.35) are given by [cf. equations (2.11) and (2.30)]

$$\begin{aligned} A_1 &= -\frac{1}{\nu_{eH}}\left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right), \\ A_2 &= -\frac{\nu_{eH}}{\nu_{eH}^2 + \omega_e^2}\left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right), \\ A_3 &= -\frac{\omega_e}{\nu_{eH}^2 + \omega_e^2}\left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right), \\ D_1 &= \frac{1}{\nu_{eH}}, \\ D_2 &= \frac{\nu_{eH}}{\nu_{eH}^2 + \omega_e^2}, \end{aligned} \tag{2.37}$$

and

$$D_3 = \frac{\omega_e}{\nu_{eH}^2 + \omega_e^2}.$$

Substitution of equations (2.35) and (2.37) for $\mathbf{f}^1$ in equations (2.31) and (2.12) for $\mathbf{q}_e$ and $\mathbf{J}_e$ then yields the following expressions for the electron heat-flux and current-density vectors:

$$\begin{aligned} \mathbf{q}_e = &- \hat{\lambda}_{\parallel} \nabla_{\parallel} T_e - \hat{\lambda}_{\perp} \nabla_{\perp} T_e - \hat{\lambda}_H \mathbf{b} \times \nabla T_e \\ &- T_e\left(\phi_{\parallel} + \frac{5}{2}\frac{k}{e}\sigma_{\parallel}\right)\mathscr{E}_{\parallel} - T_e\left(\phi_{\perp} + \frac{5}{2}\frac{k}{e}\sigma_{\perp}\right)\mathscr{E}_{\perp} \\ &- T_e\left(\phi_H + \frac{5}{2}\frac{k}{e}\sigma_H\right)\mathbf{b} \times \mathscr{E} \end{aligned} \tag{2.38}$$

and

$$\mathbf{J}_e = \sigma_{\parallel}\mathscr{E}_{\parallel} + \sigma_{\perp}\mathscr{E}_{\perp} + \sigma_H \mathbf{b} \times \mathscr{E} + \phi_{\parallel}\nabla_{\parallel} T_e + \phi_{\perp}\nabla_{\perp} T_e + \phi_H \mathbf{b} \times \nabla T_e.$$

In equations (2.38) the electron thermal and electrical conductivities $\hat{\lambda}$ and $\sigma$ are given by equations (2.34) and (2.14) as before, with the exception

that in equations (2.14) $-\partial f^0/\partial C$ is now to be replaced by $m_e C f_M/kT_e$. The *thermal-diffusion coefficients* $\phi_\parallel$, $\phi_\perp$, and $\phi_H$ are

$$\phi_\parallel = \frac{4\pi}{3}\frac{en_e}{T_e}\int_0^\infty \frac{1}{\nu_{eH}}\left(\frac{m_e C^2}{2kT_e}-\frac{5}{2}\right)C^4 f_M\, dC,$$

$$\phi_\perp = \frac{4\pi}{3}\frac{en_e}{T_e}\int_0^\infty \frac{\nu_{eH}}{\nu_{eH}^2+\omega_e^2}\left(\frac{m_e C^2}{2kT_e}-\frac{5}{2}\right)C^4 f_M\, dC, \tag{2.39}$$

and

$$\phi_H = \frac{4\pi}{3}\frac{en_e}{T_e}\int_0^\infty \frac{\omega_e}{\nu_{eH}^2+\omega_e^2}\left(\frac{m_e C^2}{2kT_e}-\frac{5}{2}\right)C^4 f_M\, dC.$$

From equations (2.39) it is easily shown that the thermal-diffusion coefficients vanish if $\nu_{eH}$ is a constant, independent of $C$. If $\nu_{eH}$ increases monotonically with $C$, $\phi_\parallel$, and $\phi_H$ are negative, whereas if $\nu_{eH}$ decreases monotonically with $C$, $\phi_\parallel$ and $\phi_H$ are positive. If $\nu_{eH}$ varies monotonically with $C$, the behavior of $\phi_\perp$ for weak magnetic fields is the same as that of $\phi_\parallel$ and $\phi_H$. For sufficiently strong magnetic fields, however, the sign of $\phi_\perp$ will be reversed.

One might suppose that as a result of thermal diffusion *two* new sets of coefficients would be required, one relating $\mathbf{q}_e$ to $\mathscr{E}$ and the other relating $\mathbf{J}_e$ to $\nabla T_e$. Direct evaluation of $\mathbf{q}_e$ and $\mathbf{J}_e$ from equations (2.35) and (2.37) shows, however, that a single set (2.39) of thermal-diffusion coefficients is sufficient. We note at this point that our use of the symbol $\hat{\lambda}$ rather than $\lambda$ for thermal conductivity is to distinguish our present form for $\mathbf{q}_e$ in terms of $\nabla T_e$ and $\mathscr{E}$ from the more usual representation of $\mathbf{q}_e$ in terms of $\nabla T_e$ and $\mathbf{J}_e$. This distinction will be discussed in greater detail in Sec. 4.

We have chosen to write $\mathbf{q}_e$ and $\mathbf{J}_e$ in terms of vector equations with scalar coefficients, such as $\sigma_\parallel$, $\sigma_\perp$, and $\sigma_H$. An alternate approach is to introduce tensor coefficients so that $\mathbf{q}_e$ and $\mathbf{J}_e$ can be expressed more compactly as

$$q_{e_\alpha} = -\hat{\lambda}_{\alpha\beta}\frac{\partial T_e}{\partial x_\beta} - T_e\left(\phi_{\alpha\beta}+\frac{5}{2}\frac{k}{e}\sigma_{\alpha\beta}\right)\mathscr{E}_\beta$$

and

$$J_{e_\alpha} = \sigma_{\alpha\beta}\mathscr{E}_\beta + \phi_{\alpha\beta}\frac{\partial T_e}{\partial x_\beta}. \tag{2.40}$$

If we adopt the coordinate system shown in Fig. 1, in which the $x_3$ direction is parallel to $\mathbf{b}$ and the orientation of the vector $\mathscr{E}$ is arbitrary, $\sigma_{\alpha\beta}$ can be expressed in terms of $\sigma_\parallel$, $\sigma_\perp$, and $\sigma_H$ by means of the matrix

$$\{\sigma_{\alpha\beta}\} = \begin{Vmatrix} \sigma_\perp & -\sigma_H & 0 \\ \sigma_H & \sigma_\perp & 0 \\ 0 & 0 & \sigma_\parallel \end{Vmatrix}. \tag{2.41}$$

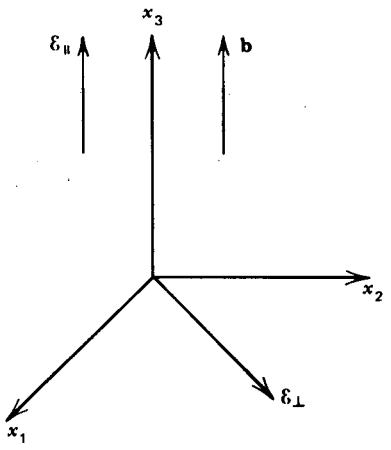



**Figure 1.** Coordinate system for the tensor electrical conductivity $\sigma_{\alpha\beta}$.

Here it is to be understood that the first subscript, $\alpha$, denotes the row of the matrix and the second subscript, $\beta$, the column. Similar relations hold for $\hat{\lambda}_{\alpha\beta}$ and $\phi_{\alpha\beta}$ (see Exercise 2.4).

As we indicated earlier, the results of this section are useful not only as they apply directly to weakly ionized plasmas, but also in that the present description remains qualitatively valid for partially ionized plasmas as well. Before discussing the theory for partially ionized plasmas, we first establish the conditions for which $f^0$ is Maxwellian.

**Exercise 2.1.** Obtain simplified expressions for the various transport coefficients (2.14), (2.34), and (2.39) that apply when $\nu_{eh}^{(1)}$ is taken to be independent of $C$. Compare your results with the mean-free-path values of $\sigma$ given by equations (II 13.7b and IV 3.11).

**Exercise 2.2.** 1. Obtain the solution (2.20) for the effect of the heavy-particle diffusion velocities on $\mathbf{f}^1$.

2. Use this result to express the corresponding contributions to $\mathbf{U}_e$ in terms of integrals similar in form to equations (2.14).

3. Simplify the results of part 2 for the case where each of the collision frequencies $\nu_{eh}^{(1)}$ is taken to be independent of $C$. With the use of integration by parts, you should recover equations (2.21).

**Exercise 2.3.** Provide the missing steps in the derivation of equations (2.35) through (2.39) for $\mathbf{q}_e$, $\mathbf{J}_e$, and the various transport properties. Show that it is necessary to introduce only one set of thermal-diffusion coefficients, that given by equations (2.39).

**Exercise 2.4.** Derive expressions similar to equation (2.41) for the tensor thermal-conductivity and thermal-diffusion coefficients $\hat{\lambda}_{\alpha\beta}$ and $\phi_{\alpha\beta}$ that appear in equations (2.40).

## 3. THE CONDITIONS FOR A MAXWELLIAN DISTRIBUTION

In this section we consider the conditions under which the electron-electron collision term $\mathbb{C}_{ee}^0$ is the dominant term in the first Cartesian-tensor equation (VII 6.39), the speed-dependent energy equation

$$\frac{D}{Dt}(n_e f^0) + \frac{C}{3}\nabla(n_e \mathbf{f}^1) - \frac{en_e}{3m_e C^2}\mathbf{E}' \cdot \frac{\partial}{\partial C}(C^2 \mathbf{f}^1) - \frac{C}{3}\nabla \cdot \mathbf{u} n_e \frac{\partial f^0}{\partial C} = \sum_h \mathbb{C}_{eh}^0 + \mathbb{C}_{ee}^0. \quad (3.1)$$

Under such conditions this equation, with $\mathbb{C}_{ee}^0$ given by equation (VII 6.32), is satisfied by

$$\mathbb{C}_{ee}^0 = n_e^2 \Gamma_{ee} \frac{1}{C^2}\frac{\partial}{\partial C}\left[f^0 I_0^0 + \frac{C}{3}\frac{\partial f^0}{\partial C}(I_2^0 + J_{-1}^0)\right] = 0, \quad (3.2)$$

where the integrals $I_p^0$ and $J_p^0$ are defined by equations (VII 6.33). That is, if all other terms in equation (3.1) are small when compared with the individual parts of $\mathbb{C}_{ee}^0$, such as $(n_e^2\Gamma_{ee}/C^2)\,\partial(f^0 I_0^0)/\partial C$, then equation (3.1) can only be satisfied if $\mathbb{C}_{ee}^0$ is itself zero.

The solution to equation (3.2), as can be verified by direct substitution, is that $f^0$ is given by the Maxwellian distribution

$$f^0 = f_M = \left(\frac{m_e}{2\pi k T_e}\right)^{3/2} e^{-m_e C^2/2kT_e}. \quad (3.3)$$

Here the electron temperature $T_e$ is to be regarded as determined by the (speed-integrated) electron energy equation, which we shall discuss later in this section.

To assess the conditions of applicability of the Maxwellian distribution (3.3) we first ask when $\mathbb{C}_{ee}^0$ will dominate the other collision terms in equation (3.1). Since we shall apply the distribution (3.3) to the calculation of transport properties, we consider the magnitude of the collision terms for electrons of thermal energies, as in the previous section. From equations (3.2) and (VII 6.33), the order of magnitude of each of the terms in $\mathbb{C}_{ee}^0$ is

$$\mathbb{C}_{ee}^0 \sim n_e^2 \Gamma_{ee}(f^0)^2 \sim n_e \nu_{ee}^{(1)} f^0. \quad (3.4)$$

Here, as in Sec. 2, we have used the relations $f^0 \sim 1/\bar{C}_e^3$ and $\nu_{ee}^{(1)} \sim n_e \Gamma_{ee}/\bar{C}_e^3$ for thermal electrons. If the heavy-particle temperature $T_h$ is not much greater than $T_e$, the order of magnitude of

$$\mathbb{C}_{eh}^0 = n_e \frac{m_e}{m_h} \frac{1}{C^2} \frac{\partial}{\partial C} \left[ C^3 \left( f^0 + \frac{kT_h}{m_e} \frac{1}{C} \frac{\partial f^0}{\partial C} \right) \nu_{eh}^{(1)} \right] \tag{3.5}$$

from equation (VII 6.26) is

$$\mathbb{C}_{eh}^0 \sim n_e \frac{m_e}{m_h} \nu_{eh}^{(1)} f^0. \tag{3.6}$$

We then have for thermal electrons

$$\frac{\mathbb{C}_{eh}^0}{\mathbb{C}_{ee}^0} \sim \frac{m_e}{m_h} \frac{\nu_{eh}^{(1)}}{\nu_{ee}^{(1)}} \sim \frac{m_e}{m_h} \frac{\bar{\nu}_{eh}}{\bar{\nu}_{ee}}, \tag{3.7}$$

where we have again introduced the mean collision frequencies (2.2). [In the present case equation (2.2) would be evaluated with $f^0 = f_M$.] Thus if

$$\sum_h \frac{m_e}{m_h} \frac{\bar{\nu}_{eh}}{\bar{\nu}_{ee}} \ll 1, \tag{3.8}$$

the terms $\mathbb{C}_{eh}^0$ in equation (3.1) can be neglected in comparison with $\mathbb{C}_{ee}^0$.

Since for electron-ion collisions $\bar{\nu}_{ei}$ is of the same order as $\bar{\nu}_{ee}$, the inequality (3.8) is always satisfied for the ions. It can then be regarded, when applied to the neutral species, as a condition on the degree of ionization of the plasma. Furthermore, since $m_e/m_n$ is a small parameter, the condition (3.8) will be satisfied for only moderately ionized plasmas. (The subscript $n$ denotes a neutral species.) To facilitate numerical estimates as to whether the condition (3.8) is satisfied for electron-neutral collisions, the following approximate formulas are useful [see equation (2.2b)]:

$$\bar{\nu}_{ee} \simeq 5 n_e T_e^{-3/2} \ln \Lambda [\text{sec}^{-1}]$$

and (3.9)

$$\frac{m_e}{m_n} \frac{\bar{\nu}_{en}}{\bar{\nu}_{ee}} \simeq \frac{8 \bar{Q}_{en} T_e^2}{W_n \ln \Lambda} \cdot \frac{n_n}{n_e}.$$

Here $\bar{Q}_{en}$ is a mean electron-neutral cross section in $\text{cm}^2$, $W_n$ is the molecular weight of the neutral species, and $n_e$ and $T_e$ are expressed in $\text{cm}^{-3}$ and °K.

To examine the terms on the left-hand side of equation (3.1) we now make more precise our use of the phrase *collision-dominated plasma*. If $\tau$ and $L$

are characteristic time and length scales for macroscopic change, we refer to a plasma as being collision-dominated, with respect to the electron gas, if

$$\frac{1}{\bar{\nu}_{ee}\tau} \ll 1 \tag{3.10a}$$

and

$$\frac{\bar{C}_e}{\bar{\nu}_{ee}L} \ll 1. \tag{3.10b}$$

If the inequalities (3.10) are satisfied, it can be seen with the use of equation (3.4) and the conditions $f^1 \ll f^0$ [equation (VII 6.7)] and $u \leq \bar{C}_e$ [equation (VII 6.38)] that all terms on the left-hand side of equation (3.1) will be small when compared with $\mathbb{C}_{ee}^0$, except possibly the third term which involves the field $\mathbf{E}'$. To assess the magnitude of this term, we can use equations (VII 6.8) to estimate that

$$\frac{\mathbf{f}^1}{f^0} \sim \frac{\mathbf{U}_e}{\bar{C}_e}. \tag{3.11}$$

It follows that the term in question will be small when compared with $\mathbb{C}_{ee}^0$ if

$$\frac{e|\mathbf{U}_e \cdot \mathbf{E}'|}{m_e \bar{C}_e^2 \bar{\nu}_{ee}} = \frac{|\mathbf{J}_e \cdot \mathbf{E}'|}{n_e m_e \bar{C}_e^2 \bar{\nu}_{ee}} \ll 1. \tag{3.12}$$

With the use of conditions (3.10) and approximate formulas for the relation of $\mathbf{J}_e$ to $\mathscr{E}$ and $\nabla T_e$, consistent in form with the results of the previous section, it can be shown that the inequality (3.12) will be satisfied when

$$\left(\frac{U_e}{\bar{C}_e}\right)^2 \frac{\bar{\nu}_{eH}}{\bar{\nu}_{ee}} \ll 1. \tag{3.13}$$

From the conditions $f^1 \ll f^0$ and (3.8), it follows that sufficient conditions for the inequalities (3.12) or (3.13) are respectively

$$\frac{e|\mathbf{E}'|}{m_e \bar{C}_e \bar{\nu}_{ee}} \leq 1$$

or

$$\left(\frac{U_e}{\bar{C}_e}\right)^2 \leq \overline{\left(\frac{m_e}{m_h}\right)},$$

where $\overline{(m_e/m_h)}$ is an average value of the mass ratio. These later conditions, however, may be unduly restrictive.

When the Joulean heating $\mathbf{J}_e \cdot \mathbf{E}'$ of the electrons is balanced in the speed-integrated energy equation [equation (3.16), which follows] primarily by an energy loss in elastic collisions, the condition (3.8) insures that the

inequality (3.12) will be satisfied (see Exercise 3.1). We note also that the inequality (3.10a), which involves a characteristic time, does not exclude from our consideration high-frequency electric fields. The Joulean heating associated with such fields, however, must satisfy the inequality (3.12).

The conditions (3.8), (3.10), (VII 6.7), and (3.12) or (3.13) [and (VII (6.38)] are sufficient to insure that $\mathbb{C}_{ee}^0$ dominates in equation (3.1) and hence that $f^0$ is Maxwellian. For the case of a uniform plasma in a strong electric field, this result has been verified numerically by Viegas and Kruger (1969), who studied the transition with increasing electron density of $f^0$ from the Lorentzian form of Sec. 2 to the Maxwellian distribution (3.3).

When $f^0$ is Maxwellian, the electron temperature must be found from the electron energy equation. The energy equation can be derived from the energy moment (VII 3.10c) of the Boltzmann equation or, equivalently, by the multiplication of the Cartesian-tensor equation (3.1) by $\frac{1}{2}m_e C^2 4\pi C^2$ and integration over all values of $C$. The latter calculation, with $\mathbb{C}_{eh}^0$ given by equation (3.5), incorporates directly the simplifications in the collision terms associated with the small electron mass. Accordingly, with $f^0 = f_M$, the heavy-particle collision term in the energy equation is readily found to be

$$\sum_h \int_0^\infty \frac{1}{2} m_e C^2 4\pi C^2 \mathbb{C}_{eh}^0 \, dC = -n_e \sum_h \frac{3}{2} k(T_e - T_h) 2 \frac{m_e}{m_h} \bar{\nu}_{eh}, \tag{3.14}$$

where the collision frequency $\bar{\nu}_{eh}$ is given by

$$\bar{\nu}_{eh} = \frac{2}{3} \int_0^\infty \frac{m_e C^2}{2kTe} f_M \nu_{eh}^{(1)} 4\pi C^2 \, dC \tag{3.15}$$

[cf. equation (2.2)]. It can be seen that the average energy lost by the electrons to the heavy particles per collision is twice the product of the mass ratio $m_e/m_h$ and the difference between the mean energies of the electrons and the heavy particles. The energy coupling between the electrons and the heavy particles is accordingly weak. The electron-electron term in the integrated energy equation vanishes, in accord with the physical fact that the net collisional energy exchange of the electron gas with itself must be zero. Thus, although the term $\mathbb{C}_{ee}^0$ dominates in equation (3.1) for a fixed value of $C$, in the integrated energy equation the net contribution of $\mathbb{C}_{ee}^0$ is zero (for any $f^0$) and the other terms must be retained.

The left-hand side of the energy equation is given by equation (VII 3.10c), in agreement with the result obtained from equation (3.1). The electron energy equation itself is then

$$\frac{D}{Dt}\left(n_e \frac{3}{2} kT_e\right) + n_e \frac{5}{2} kT_e \nabla \cdot \mathbf{u} + \nabla \cdot \mathbf{q}_e - \mathbf{J}_e \cdot \mathbf{E}' = n_e \sum_h \frac{3}{2} k(T_h - T_e) 2 \frac{m_e}{m_h} \bar{\nu}_{eh}, \tag{3.16}$$

where we have neglected as small the term $-m_e \mathbf{J}_e \cdot D\mathbf{u}/Dt$ and have neglected the electron viscous stresses $\tau_e$ in comparison with $p_e$.

Cool and Zukoski (1965) have obtained from spectroscopic measurements the electron temperature of a uniform, steady plasma for various applied electric fields. In this case, with no magnetic field, the energy equation (3.16) reduces to

$$n_e \sum_h \frac{3}{2} k(T_e - T_h) 2 \frac{m_e}{m_h} \bar{\nu}_{eh} = \mathbf{J}_e \cdot \mathbf{E}$$

$$= \sigma_{\parallel} E^2 \tag{3.17}$$

where the relation $\mathbf{J}_e = \sigma_{\parallel} \mathbf{E}$ has been used to eliminate $\mathbf{J}_e$. Fig. 2 shows a comparison of the measured values of $T_e$ in comparison with those predicted by equation (3.17), with $\sigma_{\parallel}/n_e$ calculated from the Frost formula discussed in Sec. 7. It can be seen that above fields of about 3 V/cm the agreement is good. For lower fields, however, it is found that the escape of radiation from the plasma becomes relatively more important in comparison with the Joulean heating $\sigma_{\parallel} E^2$. This radiation escape results in a loss

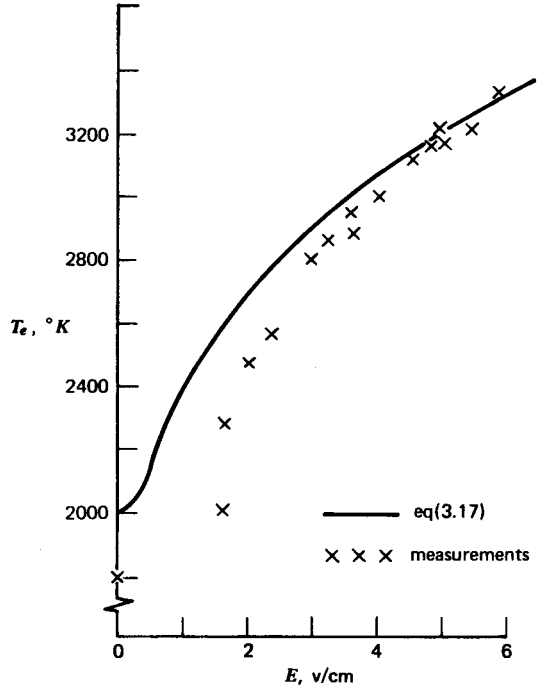



**Figure 2.** Comparison of the electron temperature as predicted by equation (3.17) with the measurements of Cool and Zukoski (1965) for atmospheric pressure potassium-seeded argon. ($n_K/n_A = 0.004$; $T_A = 2000$°K).

in energy from the electrons through nonelastic collisions and a lowering of $T_e$ below that predicted by equation (3.17). (At the same time, the escape of radiation causes the value of $T_e$ inferred from the spectroscopic measurements to fall somewhat below the actual value.) In the present discussion we have not considered the effects of nonelastic collisions. It is shown in Sec. 8 that as a result of such collisions an additional condition is required in order that $f^0 = f_M$ and that an additional, nonelastic-collision term must be added to the energy equation (3.16).

In this section, with nonequilibrium situations such as that of Cool and Zukoski in mind, we have emphasized the role of electron-electron collisions in establishing for $f^0$ a Maxwellian distribution at the electron temperature. It should also be noted that for small departures from equilibrium electron/heavy-particle collisions tend to produce a Maxwellian electron distribution with $T_e = T_h$. Thus even for low degrees of ionization for which the condition (3.8) $[\sum_h m_e \bar{\nu}_{eh}/m_h \bar{\nu}_{ee} \ll 1]$ is not satisfied, $f^0$ can still be Maxwellian at $T_h$, as is shown for small $\mathbf{E}'$ by equation (2.18).

The situation for a uniform, steady plasma for which an electric field is the driving force for nonequilibrium can be summarized by means of the diagram shown in Fig. 3. On this diagram the regions in which the results of this and the preceding section apply are indicated approximately in terms of the electric-field strength (the ordinate) and the degree of ionization

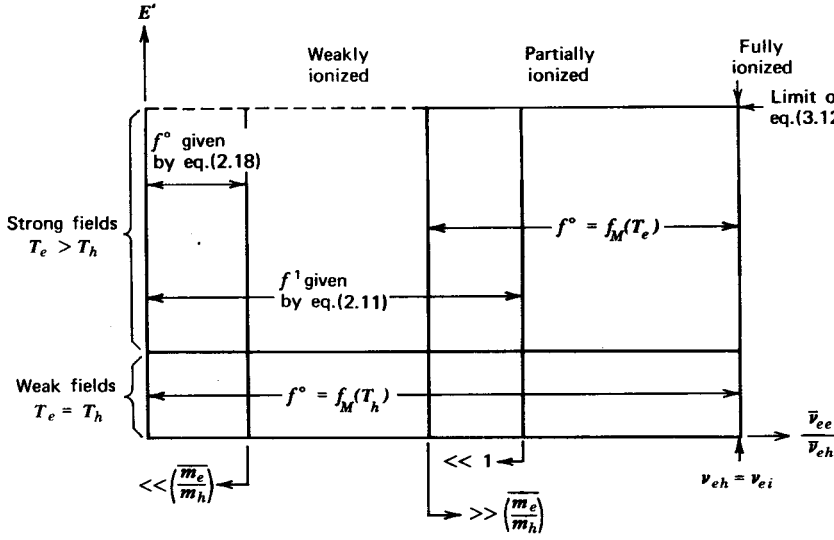



Figure 3. Conditions for a Maxwellian distribution [after Viegas and Kruger (1969)].

expressed as $\bar{\nu}_{ee}/\bar{\nu}_{eh}$ (the abscissa). The approximate boundary on the ordinate between regions of weak and strong fields can be found with the use of equation (3.17). The classical Chapman-Enskog theory of transport properties applies strictly only in the weak-field region where $T_e = T_h$ and $f^0 = f_M(T_h)$. As we shall find in the following section, however, by means of the Cartesian-tensor expansion it is possible to extend the rigorous theory of transport properties to the entire region where $f^0$ is Maxwellian.

**Exercise 3.1.** Show that the condition (3.8) insures that the inequality (3.12) will always be satisfied in those situations for which the energy equation is given by equation (3.17).

**Exercise 3.2.** Derive from equation (3.5) the collision term (3.14) in the electron energy equation (3.16), and show that equation (3.16) follows from equation (VII 3.10c).

**Exercise 3.3.** With the use of the Saha equation (II 10.5b), find the approximate range in temperature and pressure for which the inequality (3.8) is satisfied for equilibrium partially ionized argon.

## 4. PARTIALLY IONIZED PLASMAS

The Chapman-Enskog theory of the transport properties of a gas mixture is discussed by Chapman and Cowling (1952) and by Hirschfelder, Curtiss and Bird (1964). In this theory it is assumed that the driving mechanisms for translational nonequilibrium of the gas are such that the departures of the distribution functions of the various species from a Maxwellian distribution are small. Such conditions are in fact the microscopic justification for the description of the nonequilibrium phenomena of viscous stress, heat conduction, and diffusion in terms of transport properties. As a consequence of these conditions, it is found in the classical theory that the temperatures of the species remain equal in a nonequilibrium situation. In the preceding section, we have seen that similar limitations on the degree of nonequilibrium [equations (3.10) and (3.12)] are necessary in order that the isotropic part of the electron distribution function in a plasma be Maxwellian. As a result of the small electron mass, however, the electron temperature as given by the electron energy equation can depart significantly from the heavy-particle temperature.

In this section, we consider nonequilibrium situations where the conditions of Sec. 3 apply and discuss the solution for the anisotropic part $\mathbf{f}^1$ of the electron distribution function. This solution yields expressions for the electron current density and heat-flux vector in the presence of a magnetic

field in terms of transport coefficients. These coefficients can be calculated from the theory to varying degrees of approximation. On the basis of mean-free-path considerations, the electron viscosity coefficient $\eta_e$, which depends on the solution for $f_{ij}^2$, is of the order of

$$\frac{\eta_e}{\eta_h} \sim \frac{n_e}{n_h}\left(\frac{m_e T_e}{m_h T_h}\right)^{1/2}$$

when compared with the heavy-particle viscosity $\eta_h$. Since the electron contribution to the viscous stress is therefore almost always negligible, we shall not concern ourselves with the calculation of $\eta_e$.

As we have already indicated, our emphasis on electron transport properties stems from the special role of electrons in plasma transport phenomena, particularly in the presence of a magnetic field, and the particular simplifications that are possible in the calculations of these properties. For example, in many situations the current in a plasma is carried almost entirely by the electrons. Furthermore, as we saw in Chapter VII, the electron Boltzmann equation can be decoupled from the heavy-particle Boltzmann equations, so that the solution for the electron transport properties can be considered separately. The consequences of this decoupling on the calculation of the heavy-particle transport properties will be discussed briefly in Sec. 6.

When the conditions of Sec. 3 are satisfied, the isotropic part $f^0$ of the electron distribution is Maxwellian and the function $\mathbf{f}^1$ is governed solely by equation (VII 6.40). We shall not consider at present high-frequency electric fields and, on the basis of the condition $U_h \ll U_e$ discussed in Sec. 2, shall neglect the part of the collision term in equation (VII 6.40) that involves $\mathbf{U}_h$. Equation (VII 6.40) with the time derivative omitted then becomes

$$f_M C\left[\left(\frac{5}{2} - \mathscr{C}^2\right)\nabla \ln T_e - \frac{e}{kT_e}\boldsymbol{\mathscr{E}}\right] = -\boldsymbol{\omega}_e \times \mathbf{f}^1 + \nu_{eH}\mathbf{f}^1 - \frac{1}{n_e}\mathbf{C}_{ee}^1, \quad (4.1)$$

where $\mathbf{C}_{ee}^1$ is now given by the use of $f^0 = f_M$ in equation (VII 6.36). In writing equation (4.1) we have used the relation [cf. equation (2.32)]

$$\begin{aligned}\frac{1}{n_e} C\nabla n_e f_M &= C f_M\left[\nabla \ln n_e + \left(\frac{m_e C^2}{2kT_e} - \frac{3}{2}\right)\nabla \ln T_e\right] \\ &= C f_M\left[\nabla \ln p_e + \left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right)\nabla \ln T_e\right]\end{aligned}$$

and the following notation:

$$\mathscr{C}^2 = \frac{m_e C^2}{2kT_e},$$
$$\omega_e \equiv e\mathbf{B}/m_e, \tag{4.2}$$

and, as before,

$$\mathscr{E} = \mathbf{E}' + \frac{\nabla p_e}{en_e} + \frac{m_e}{e}\frac{D\mathbf{u}}{Dt}$$
$$\simeq \mathbf{E}' + \frac{\nabla p_e}{en_e}. \tag{4.3}$$

As in Sec. 2, the generalized electric field $\mathscr{E}$ contains in $\nabla p_e/en_e$ a part of the term $C\nabla n_e f_M$. Again, as indicated, the last term in $\mathscr{E}$ can be neglected in almost all practical situations.

Equation (4.1) is a linear equation for $\mathbf{f}^1$, since $\mathbb{C}_{ee}^1$ is a linear function of $\mathbf{f}^1$. Furthermore, the driving terms $\nabla \ln T_e$ and $\mathscr{E}$ for translational nonequilibrium appear linearly and are mutually independent. Accordingly, the solution of equation (4.1) can be written in the form [cf. equation (2.35)]

$$\mathbf{f}^1 = f_M C[A_1 \nabla_\parallel \ln T_e + A_2 \nabla_\perp \ln T_e + A_3 \mathbf{b} \times \nabla \ln T_e - (e/kT_e)(D_1 \mathscr{E}_\parallel + D_2 \mathscr{E}_\perp + D_3 \mathbf{b} \times \mathscr{E})], \tag{4.4}$$

where we have introduced the factor $-e/kT_e$ for later convenience. As in Sec. 2, the subscripts $\parallel$ and $\perp$ refer to components parallel and perpendicular to the magnetic field. Equations for the unknown scalar functions $A_1$, $A_2$, $A_3$, $D_1$, $D_2$, and $D_3$ of electron speed are obtained by substituting equation (4.4) in equation (4.1) and equating coefficients in the three mutually perpendicular directions $(\ )_\parallel$, $(\ )_\perp$, and $\mathbf{b} \times (\ )$ for both $\nabla \ln T_e$ and $\mathscr{E}$. Using the fact that $\mathbb{C}_{ee}^1$ is a linear function of $\mathbf{f}^1$, we thus obtain the equations

$$\begin{aligned}
f_M C\left(\frac{5}{2} - \mathscr{C}^2\right) &= + \nu_{eH} f_M C A_1 - \mathbb{C}_{ee}^1[f_M C A_1]/n_e \\
f_M C\left(\frac{5}{2} - \mathscr{C}^2\right) &= \omega_e f_M C A_3 + \nu_{eH} f_M C A_2 - \mathbb{C}_{ee}^1[f_M C A_2]/n_e \\
0 &= -\omega_e f_M C A_2 + \nu_{eH} f_M C A_3 - \mathbb{C}_{ee}^1[f_M C A_3]/n_e \\
f_M C &= + \nu_{eH} f_M C D_1 - \mathbb{C}_{ee}^1[f_M C D_1]/n_e \\
f_M C &= \omega_e f_M C D_3 + \nu_{eH} f_M C D_2 - \mathbb{C}_{ee}^1[f_M C D_2]/n_e \\
0 &= -\omega_e f_M C D_2 + \nu_{eH} f_M C D_3 - \mathbb{C}_{ee}^1[f_M C D_3]/n_e.
\end{aligned} \tag{4.5}$$

Here $\mathbb{C}^1_{ee}[\phi]$ denotes the linear, scalar function of $\phi$ that is obtained from equation (VII 6.36) for $\mathbb{C}^1_{ee}$ by the replacement of $f^0$ by $f_M$ and $\mathbf{f}^1$ by $\phi$ in that equation. That is, $\mathbb{C}^1_{ee}[\phi]$ is given by

$$\mathbb{C}^1_{ee}[\phi] = n_e^2 \Gamma_{ee}\left[8\pi f_M \phi + \frac{1}{15C^2}\frac{\partial f_M}{\partial C}(-3I_3^\phi + 5I_1^\phi + 2J_{-2}^\phi)\right.$$
$$+ \frac{1}{5C}\frac{\partial^2 f_M}{\partial C^2}(I_3^\phi + J_{-2}^\phi) + \frac{1}{3C^2}(-I_2^M + 3I_0^M + 2J_{-1}^M)\left(-\frac{\phi}{C} + \frac{\partial \phi}{\partial C}\right)$$
$$\left. + \frac{1}{3C}(I_2^M + J_{-1}^M)\frac{\partial^2 \phi}{\partial C^2}\right], \tag{4.6}$$

where

$$I_p^\phi = \frac{4\pi}{C^p}\int_0^C C^{p+2}\phi \, dC,$$

$$J_p^\phi = \frac{4\pi}{C^p}\int_C^\infty C^{p+2}\phi \, dC,$$

$$I_p^M = \frac{4\pi}{C^p}\int_0^C C^{p+2} f_M \, dC,$$

and

$$J_p^M = \frac{4\pi}{C^p}\int_C^\infty C^{p+2} f_M \, dC.$$

We note from equations (4.5) that the equations for $A_1$ and $D_1$ are independent of $\omega_e$, or equivalently $B$, and that the equations for $A_2$ and $D_2$ reduce to those for $A_1$ and $D_1$ when $B = 0$. It follows that $A_1 = A_2(B = 0)$ and that $D_1 = D_2(B = 0)$. We can therefore limit our consideration to the solutions for $A_2$, $A_3$, $D_2$, and $D_3$ and recover $A_1$ and $D_1$ by setting $B = 0$ in the solutions for $A_2$ and $D_2$. The solutions for these quantities can be most conveniently obtained in terms of the complex variables $A$ and $D$ that are defined by

$$A \equiv A_2 + iA_3 \tag{4.7}$$

and

$$D \equiv D_2 + iD_3$$

where $i^2 = -1$. Equations (4.5) then reduce to

$$f_M \mathscr{C}(\tfrac{5}{2} - \mathscr{C}^2) = (\nu_{eH} - i\omega_e) f_M \mathscr{C} A - \mathbb{C}^1_{ee}[f_M \mathscr{C} A]/n_e \tag{4.8}$$

and

$$f_M \mathscr{C} = (\nu_{eH} - i\omega_e) f_M \mathscr{C} D - \mathbb{C}^1_{ee}[f_M \mathscr{C} D]/n_e. \tag{4.9}$$

The solutions of Sec. 2 for a Lorentzian plasma, with $f^0 = f_M$, can be recovered directly from equations (4.8) and (4.9) by setting $\mathbb{C}_{ee}^1$ equal to zero in these equations.

Before proceeding with the solution of equations (4.8) and (4.9) for $A$ and $D$, we examine the consequences of the use of the form (4.4) for $\mathbf{f}^1$ in the expressions (2.12) and (2.31) for the electron current density $\mathbf{J}_e$ and heat flux $\mathbf{q}_e$. These expressions are

$$\mathbf{J}_e = -en_e \frac{4\pi}{3} \int_0^\infty C^3 \mathbf{f}^1 \, dC \tag{4.10}$$

and

$$\mathbf{q}_e = n_e m_e \frac{2\pi}{3} \int_0^\infty C^5 \mathbf{f}^1 \, dC = -\frac{5}{2} \frac{kT_e}{e} \mathbf{J}_e + n_e m_e \frac{2\pi}{3} \int_0^\infty \left( C^2 - \frac{5kT_e}{m_e} \right) C^3 \mathbf{f}^1 \, dC,$$

where we have written the second form for $\mathbf{q}_e$ to simplify our later calculations. With the use of equation (4.4) for $\mathbf{f}^1$, $\mathbf{J}_e$ and $\mathbf{q}_e$ can be written directly in terms of transport coefficients as

$$\mathbf{J}_e = \sigma_\parallel \mathscr{E}_\parallel + \sigma_\perp \mathscr{E}_\perp + \sigma_H \mathbf{b} \times \mathscr{E} + \phi_\parallel \nabla_\parallel T_e + \phi_\perp \nabla_\perp T_e + \phi_H \mathbf{b} \times \nabla T_e, \tag{4.11}$$

and

$$\mathbf{q}_e = -\frac{5}{2} \frac{kT_e}{e} \mathbf{J}_e - \lambda'_\parallel \nabla_\parallel T_e - \lambda'_\perp \nabla_\perp T_e - \lambda'_H \mathbf{b} \times \nabla T_e$$

$$- T_e \phi'_\parallel \mathscr{E}_\parallel - T_e \phi'_\perp \mathscr{E}_\perp - T_e \phi'_H \mathbf{b} \times \mathscr{E}. \tag{4.12}$$

The coefficients of electrical conductivity, $\sigma$, thermal conductivity, $\lambda'$, and thermal diffusion $\phi$ and $\phi'$ are given by

$$\sigma_\parallel = \frac{n_e e^2}{kT_e} \frac{4\pi}{3} \int_0^\infty C^4 D_1 f_M \, dC = \frac{n_e e^2}{m_e} \frac{8}{3\sqrt{\pi}} \int_0^\infty \mathscr{C}^4 D_1 e^{-\mathscr{C}^2} \, d\mathscr{C},$$

$$\sigma_\perp = \frac{n_e e^2}{m_e} \frac{8}{3\sqrt{\pi}} \int_0^\infty \mathscr{C}^4 D_2 e^{-\mathscr{C}^2} \, d\mathscr{C},$$

$$\sigma_H = \frac{n_e e^2}{m_e} \frac{8}{3\sqrt{\pi}} \int_0^\infty \mathscr{C}^4 D_3 e^{-\mathscr{C}^2} \, d\mathscr{C},$$

$$\phi_\parallel = \frac{-n_e ek}{m_e} \frac{8}{3\sqrt{\pi}} \int_0^\infty \mathscr{C}^4 A_1 e^{-\mathscr{C}^2} \, d\mathscr{C},$$

$$\phi_\perp = \frac{-n_e ek}{m_e}\frac{8}{3\sqrt{\pi}}\int_0^\infty \mathscr{C}^4 A_2 e^{-\mathscr{C}^2}\, d\mathscr{C}$$

$$\phi_H = \frac{-n_e ek}{m_e}\frac{8}{3\sqrt{\pi}}\int_0^\infty \mathscr{C}^4 A_3 e^{-\mathscr{C}^2}\, d\mathscr{C},$$

$$\lambda'_\parallel = \frac{-n_e k^2 T_e}{m_e}\frac{8}{3\sqrt{\pi}}\int_0^\infty \left(\mathscr{C}^2 - \frac{5}{2}\right)\mathscr{C}^4 A_1 e^{-\mathscr{C}^2}\, d\mathscr{C},$$

$$\lambda'_\perp = \frac{-n_e k^2 T_e}{m_e}\frac{8}{3\sqrt{\pi}}\int_0^\infty \left(\mathscr{C}^2 - \frac{5}{2}\right)\mathscr{C}^4 A_2 e^{-\mathscr{C}^2}\, d\mathscr{C},$$

$$\lambda'_H = \frac{-n_e k^2 T_e}{m_e}\frac{8}{3\sqrt{\pi}}\int_0^\infty \left(\mathscr{C}^2 - \frac{5}{2}\right)\mathscr{C}^4 A_3 e^{-\mathscr{C}^2}\, d\mathscr{C}, \tag{4.13}$$

$$\phi'_\parallel = \frac{n_e ek}{m_e}\frac{8}{3\sqrt{\pi}}\int_0^\infty \left(\mathscr{C}^2 - \frac{5}{2}\right)\mathscr{C}^4 D_1 e^{-\mathscr{C}^2}\, d\mathscr{C},$$

$$\phi'_\perp = \frac{n_e ek}{m_e}\frac{8}{3\sqrt{\pi}}\int_0^\infty \left(\mathscr{C}^2 - \frac{5}{2}\right)\mathscr{C}^4 D_2 e^{-\mathscr{C}^2}\, d\mathscr{C},$$

and

$$\phi'_H = \frac{n_e ek}{m_e}\frac{8}{3\sqrt{\pi}}\int_0^\infty \left(\mathscr{C}^2 - \frac{5}{2}\right)\mathscr{C}^4 D_3 e^{-\mathscr{C}^2}\, d\mathscr{C}.$$

Although the expressions (4.13) when taken together are complicated in appearance, their derivation from equations (4.4) and (4.10) is completely straightforward. As in Sec. 2, the subscript $H$ denotes the Hall direction, which is mutually perpendicular to the magnetic field and to the driving "force," $\mathscr{E}$ or $\nabla T_e$. We shall find from the following analysis that the two thermal-diffusion coefficients $\phi$ and $\phi'$ are actually the same; that is

$$\phi'_\parallel = \phi_\parallel$$
$$\phi'_\perp = \phi_\perp$$
$$\phi'_H = \phi_H.$$

The coefficients $\lambda'$ differ from the usual coefficients of thermal conductivity $\lambda$. The coefficients $\lambda$ are such that the heat flux is proportional to $\lambda$ when the current density is zero. In terms of $\lambda$, equation (4.12) for the heat flux can be rewritten

$$\mathbf{q}_e = -\lambda_\parallel \nabla_\parallel T_e - \lambda_\perp \nabla_\perp T_e - \lambda_H \mathbf{b} \times \nabla T_e + \psi_\parallel \mathbf{J}_{e\parallel} + \psi_\perp \mathbf{J}_{e\perp} + \psi_H \mathbf{b} \times \mathbf{J}_e, \tag{4.14}$$

where the $\psi$'s are new coefficients.

The coefficients $\lambda$ and $\psi$ can be found with the aid of equation (4.11). The result is (see Marshall, 1960)

$$\begin{aligned}
\lambda_{\parallel} &= \lambda'_{\parallel} - T_e \phi_{\parallel}^2/\sigma_{\parallel} \\
\lambda_{\perp} &= \lambda'_{\perp} - T_e \frac{\sigma_{\perp}(\phi_{\perp}^2 - \phi_H^2) + 2\sigma_H \phi_{\perp} \phi_H}{\sigma_{\perp}^2 + \sigma_H^2}, \\
\lambda_H &= \lambda'_H - T_e \frac{\sigma_H(\phi_H^2 - \phi_{\perp}^2) + 2\sigma_{\perp} \phi_{\perp} \phi_H}{\sigma_{\perp}^2 + \sigma_H^2},
\end{aligned} \tag{4.15}$$

$$\begin{aligned}
\psi_{\parallel} &= -\frac{5}{2}\frac{kT_e}{e} - T_e \frac{\phi_{\parallel}}{\sigma_{\parallel}}, \\
\psi_{\perp} &= -\frac{5}{2}\frac{kT_e}{e} - T_e \frac{\sigma_{\perp}\phi_{\perp} + \sigma_H \phi_H}{\sigma_{\perp}^2 + \sigma_H^2},
\end{aligned}$$

and

$$\psi_H = -T_e \frac{\sigma_{\perp}\phi_H - \sigma_H \phi_{\perp}}{\sigma_{\perp}^2 + \sigma_H^2}.$$

Here we have used the previously mentioned fact that $\phi' = \phi$.

It is also possible to express the heat flux $\mathbf{q}_e$ directly in terms of $\nabla T_e$ and $\mathscr{E}$, with the elimination of $\mathbf{J}_e$ from equation (4.12) by means of equation (4.11). This is the form that we used in Sec. 2, with the coefficients of $\nabla T_e$ in that expression denoted by $\hat{\lambda}$ [see equation (2.38)]. Our present choice of the form (4.12) for $\mathbf{q}_e$ is motivated by the fact that for a partially ionized plasma the coefficients $\sigma$, $\lambda'$, and $\phi$ can be calculated most readily. Once these quantities are known, the coefficients $\lambda$ and $\psi$ can be found from equations (4.15). In the absence of a magnetic field, the foregoing relations reduce to the following forms:

$$\mathbf{J}_e = \sigma_{\parallel} \mathscr{E} + \phi_{\parallel} \nabla T_e$$

and

$$\begin{aligned}
\mathbf{q}_e &= -\frac{5}{2}\frac{kT_e}{e}\mathbf{J}_e - \lambda'_{\parallel} \nabla T_e - T_e \phi_{\parallel} \mathscr{E} \\
&= -\left(\lambda'_{\parallel} + \frac{5}{2}\frac{kT_e}{e}\phi_{\parallel}\right)\nabla T_e - T_e\left(\phi_{\parallel} + \frac{5}{2}\frac{k}{e}\sigma_{\parallel}\right)\mathscr{E}
\end{aligned} \tag{4.16}$$

or

$$\mathbf{q}_e = -\lambda_{\parallel} \nabla T_e + \psi_{\parallel} \mathbf{J}_e.$$

At this point we have established the functional dependence of the fluxes $\mathbf{J}_e$ and $\mathbf{q}_e$ on the driving forces $\mathscr{E}$ and $\nabla T_e$. We now consider the calculation of the transport properties themselves. As in the kinetic theory of gas mixtures, although a closed-form solution for these properties is not possible, an approximate solution can be obtained by means of expansion in terms of Sonine polynomials. The Sonine polynomials $S_m^{(n)}$ are defined by

$$S_m^{(n)}[x] = \sum_{p=0,1,2,\ldots}^{n} \frac{(m+n)!}{(m+p)!\,p!\,(n-p)!}(-x)^p \tag{4.17a}$$

so that, for example,

$$S_m^{(0)}[x] = 1 \tag{4.17b}$$

and

$$S_m^{(1)}[x] = m + 1 - x. \tag{4.17c}$$

These polynomials have the orthogonality property

$$\int_0^\infty x^m e^{-x} S_m^{(n)}[x] S_m^{(q)}[x]\,dx = 0 \qquad \text{when } n \neq q$$

$$= \frac{(m+n)!}{n!} \qquad \text{when } n = q. \tag{4.18}$$

In this relation the factorial of a number other than an integer is to be interpreted as a $\Gamma$-function. In particular, if $q$ is an integer, we have

$$\left(\frac{q}{2}\right)! = \left(\frac{q}{2}\right)\left(\frac{q}{2} - 1\right)\cdots\frac{3}{2}\frac{1}{2}\sqrt{\pi}. \tag{4.19}$$

To obtain approximate solutions to equations (4.8) and (4.9) for the quantities $A$ and $D$, we write

$$A \simeq A^{(\xi)} \equiv \sum_{r=1}^{\xi} \alpha_r^{(\xi)} S_{3/2}^{(r-1)}[\mathscr{C}^2] \tag{4.20}$$

and

$$D \simeq D^{(\xi)} \equiv \sum_{r=1}^{\xi} \delta_r^{(\xi)} S_{3/2}^{(r-1)}[\mathscr{C}^2]. \tag{4.21}$$

The expansion coefficients $\alpha_r^{(\xi)}$ and $\delta_r^{(\xi)}$ depend on the order $\xi$ of the approximation. Since the quantities $A$ and $D$ are complex [see equations

(4.7)], $\alpha_r^{(\xi)}$ and $\delta_r^{(\xi)}$ are also complex. The transport properties $\sigma$, $\lambda'$, and $\phi$ can be obtained in terms of the expansion coefficients $\alpha_r^{(\xi)}$ and $\delta_r^{(\xi)}$ by substitution of the expansions (4.20) and (4.21) in equations (4.13). With the use of the orthogonality property (4.18) and equations (4.17), we find that

$$\begin{aligned}
\sigma_\perp^{(\xi)} + i\sigma_H^{(\xi)} &= \frac{n_e e^2}{m_e}\delta_1^{(\xi)}, \\
\phi_\perp^{(\xi)} + i\phi_H^{(\xi)} &= -\frac{n_e ek}{m_e}\alpha_1^{(\xi)}, \\
\lambda_\perp'^{(\xi)} + i\lambda_H'^{(\xi)} &= \frac{5n_e k^2 T_e}{2m_e}\alpha_2^{(\xi)},
\end{aligned} \tag{4.22}$$

and

$$\phi_\perp'^{(\xi)} + i\phi_H'^{(\xi)} = -\frac{5}{2}\frac{n_e ek}{m_e}\delta_2^{(\xi)}.$$

(The proof that $\phi' = \phi$, which follows, is based on these relations.) Since $A_1 = A_2(B = 0)$ and $D_1 = D_2(B = 0)$, we have also

$$\begin{aligned}
\sigma_\parallel^{(\xi)} &= \sigma_\perp^{(\xi)}(B = 0), \\
\phi_\parallel^{(\xi)} &= \phi_\perp^{(\xi)}(B = 0), \\
\lambda_\parallel'^{(\xi)} &= \lambda_\perp'^{(\xi)}(B = 0), \\
\phi_\parallel'^{(\xi)} &= \phi_\perp'^{(\xi)}(B = 0).
\end{aligned} \tag{4.23}$$

The expansion coefficients are subject to the governing equations (4.8) and (4.9) for $A$ and $D$. These equations can be written in terms of Sonine polynomials as

$$\left(\frac{m_e}{2\pi kT_e}\right)^{3/2} e^{-\mathscr{C}^2}\mathscr{C}S_{3/2}^{(1)}[\mathscr{C}^2] = (\nu_{eH} - i\omega_e)\mathscr{C}f_M A - \mathbb{C}_{ee}^1[\mathscr{C}f_M A]/n_e$$

and (4.24)

$$\left(\frac{m_e}{2\pi kT_e}\right)^{3/2} e^{-\mathscr{C}^2}\mathscr{C}S_{3/2}^{(0)}[\mathscr{C}^2] = (\nu_{eH} - i\omega_e)\mathscr{C}f_M D - \mathbb{C}_{ee}^1[\mathscr{C}f_M D]/n_e.$$

The approximate solutions to equations (4.24) are obtained from moment equations that are formed by the substitution in these equations of the

expansions (4.20) and (4.21) for $A$ and $D$, followed by the successive multiplication of the resulting equations by $\mathscr{C}S_{3/2}^{(s-1)}[\mathscr{C}^2]4\pi C^2\,dC$ ($s = 1, 2, 3, \ldots, \xi$) and integration over $C$. These integrations are readily performed with the use of the orthogonality property (4.18) and yield the following set of algebraic equations for the expansion coefficients $\alpha_r^{(\xi)}$ and $\delta_r^{(\xi)}$:

$$\sum_{r=1}^{\xi} \alpha_r^{(\xi)} a_{rs} = \begin{cases} \frac{15}{4} & \text{if } s = 2 \\ 0 & \text{if } s \neq 2 \end{cases} \tag{4.25}$$

and

$$\sum_{r=1}^{\xi} \delta_r^{(\xi)} a_{rs} = \begin{cases} \frac{3}{2} & \text{if } s = 1 \\ 0 & \text{if } s \neq 1 \end{cases}. \tag{4.26}$$

Since $s = 1, 2, 3, \ldots, \xi$, equations (4.25) and (4.26) represent $\xi$ equations for each of the $\xi$ unknown coefficients $\alpha_r^{(\xi)}$ and $\delta_r^{(\xi)}$. The quantities $a_{rs}$ are defined by the relations

$$a_{rs} \equiv -i\omega_e \frac{2}{\sqrt{\pi}} \frac{(r+\frac{1}{2})!}{(r-1)!} \delta_{rs} + a_{rs}^{(h)} + a_{rs}^{(e)}, \tag{4.27}$$

$$a_{rs}^{(h)} \equiv \int_0^\infty \mathscr{C}^2 S_{3/2}^{(r-1)}[\mathscr{C}^2] S_{3/2}^{(s-1)}[\mathscr{C}^2] \nu_{eH} f_M 4\pi C^2\,dC, \tag{4.28}$$

and

$$a_{rs}^{(e)} \equiv -\frac{1}{n_e} \int_0^\infty S_{3/2}^{(s-1)}[\mathscr{C}^2] \mathscr{C} \mathbb{C}_{ee}^1 \left[ f_M \mathscr{C} S_{3/2}^{(r-1)}[\mathscr{C}^2] \right] 4\pi C^2\,dC, \tag{4.29}$$

where $\delta_{rs}$ is the Kronecker delta.

The definite integrals (4.29) for $a_{rs}^{(e)}$ involve only the electron-electron collision frequency and hence can be evaluated once and for all. The pertinent calculations have been performed by several authors (e.g., Landshoff, 1949). The results can be expressed in the form

$$a_{rs}^{(e)} = \frac{3\sqrt{2}}{4} \bar{\nu}_{ee} a'_{rs}, \tag{4.30}$$

in terms of the mean electron-electron collision frequency (2.2b)

$$\bar{\nu}_{ee} = n_e \frac{8\sqrt{\pi}}{3} \left( \frac{m_e}{kT_e} \right)^{3/2} \left( \frac{e^2}{4\pi\epsilon_0 m_e} \right)^2 \ln \Lambda. \tag{4.31}$$

The numbers $a'_{rs} = a'_{sr}$ are found to satisfy the identity

$$-\sum_{r,s} \zeta^{r-1}\eta^{s-1}a'_{rs} \equiv \frac{\zeta\eta(-8 + 4\zeta + 4\eta + \zeta\eta - 2\zeta^2\eta - 2\zeta\eta^2 + 3\zeta^2\eta^2)}{(1-\zeta\eta)^2(2-\zeta-\eta)^{5/2}}. \quad (4.32)$$

That is, the numbers $a'_{rs}$ are equal to the expansion coefficients of the expression on the right-hand side of equation (4.32). Daybelge (1968) has evaluated these quantities up to $r = s = 12$. The first few values are

$$a'_{rs} = \sqrt{2}\begin{Bmatrix} 0 & 0 & 0 & 0 & 0 & \cdot \\ 0 & 1 & .75000 & .46875 & .27344 & \cdot \\ 0 & .75000 & 2.8125 & 2.41406 & 1.72852 & \cdot \\ 0 & .46875 & 2.41406 & 5.52441 & 4.96802 & \cdot \\ 0 & .27344 & 1.72852 & 4.96802 & 9.13995 & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{Bmatrix} \quad (4.33)$$

These values have been calculated on the basis of a "cut-off" Coulomb potential and hence apply when $\ln \Lambda \gg 1$ (see Chapter VII, Sec. 4). For smaller $\ln \Lambda$, Daybelge (1968) and (1970) gives corrections based on the unified theory of Kihara and Aona (1967). From the standpoint of practical calculations, however, the correction for smaller $\ln \Lambda$ may be most readily applied by means of the Frost mixture rule discussed in Sec. 7.

The integrals $a_{rs}^{(h)}$ depend on the collision frequency $\nu_{eH} = \sum_h \nu_{eh}^{(1)}$ and must be evaluated for the particular plasma conditions in question. The contributions of the various species to $a_{rs}^{(h)}$, when divided by $n_h$, are identical to the small-electron-mass approximations to the so-called bracket integrals that appear in the Chapman-Enskog theory. In the present case, however, the considerations of the dynamics of a collision have already been incorporated in $a_{rs}^{(h)}$. Thus the evaluation of the integrals $a_{rs}^{(h)}$ from equation (4.28) and given collision cross sections reduces to a matter of straightforward quadratures.

Once the quantities $a_{rs}$ have been calculated, the transport properties $\sigma$, $\lambda'$, and $\phi$ can be found from the solutions of the equations (4.25) and (4.26) for $\delta_1^{(\xi)}, \alpha_1^{(\xi)}, \alpha_2^{(\xi)}$, and $\delta_2^{(\xi)}$ by the use of equations (4.22) and (4.23). The solutions to equations (4.25) and (4.26) can be expressed in terms of determinants by means of Cramer's rule. The values of the transport properties calculated from these equations will depend on the order $\xi$ of the approximation and correspond directly to the $\xi$th approximations in the classical Chapman-Enskog theory. It follows from equations (4.22), (4.23), (4.25), and (4.26) that nonzero values of $\sigma$ are obtained for $\xi \geq 1$, whereas the calculation of $\lambda'$ and $\phi$ requires $\xi \geq 2$.

To prove that the two thermal diffusion coefficients $\phi'$ and $\phi$ are

actually the same, we write the solutions to equations (4.25) and (4.26) in terms of determinants as

$$\alpha_1^{(\xi)} = \frac{\begin{vmatrix} 0 & a_{21} & \cdot & \cdot & a_{\xi 1} \\ \frac{15}{4} & a_{22} & \cdot & \cdot & a_{\xi 2} \\ 0 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & a_{2\xi} & \cdot & \cdot & a_{\xi\xi} \end{vmatrix}}{\Delta} = -\frac{15}{4} \frac{\begin{vmatrix} a_{21} & a_{31} & \cdot & a_{\xi 1} \\ a_{23} & a_{33} & \cdot & a_{\xi 3} \\ \cdot & \cdot & \cdot & \cdot \\ a_{2\xi} & \cdot & \cdot & a_{\xi\xi} \end{vmatrix}}{\Delta}$$

and

$$\delta_2^{(\xi)} = \frac{\begin{vmatrix} a_{11} & \frac{3}{2} & a_{31} & \cdot & a_{\xi 1} \\ a_{12} & 0 & a_{32} & \cdot & a_{\xi 2} \\ \cdot & 0 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{1\xi} & 0 & a_{3\xi} & \cdot & a_{\xi\xi} \end{vmatrix}}{\Delta} = -\frac{3}{2} \frac{\begin{vmatrix} a_{12} & a_{32} & \cdot & a_{\xi 2} \\ a_{13} & a_{33} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ a_{1\xi} & \cdot & \cdot & a_{\xi\xi} \end{vmatrix}}{\Delta},$$

where $\Delta$ is the determinant of the coefficients

$$\Delta \equiv \begin{vmatrix} a_{11} & a_{21} & \cdot & \cdot & a_{\xi 1} \\ a_{12} & a_{22} & \cdot & \cdot & a_{\xi 2} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{1\xi} & \cdot & \cdot & \cdot & a_{\xi\xi} \end{vmatrix}$$

Since $a_{rs} = a_{sr}$ and since the value of a determinant is unchanged by interchange of the rows and columns, it follows that

$$\tfrac{5}{2}\delta_2^{(\xi)} = \alpha_1^{(\xi)}, \tag{4.34}$$

for any order $\xi$ of approximation. Therefore, from equations (4.22) and (4.23) we have for each component $\phi' = \phi$, as we have previously indicated.

The preceding description of electron transport phenomena in partially ionized plasmas applies whether or not the electron temperature $T_e$ is equal to the heavy-particle temperature $T_h$, as long as the conditions of Sec. 3 are satisfied. Our results are in fact consistent with those obtained from the Chapman-Enskog theory by formal replacement of the gas temperature by the electron temperature. The present formulas are, however, much simpler in that they incorporate explicitly the simplifications that result from the

small electron mass. Accordingly, they provide a simplified basis for the calculation of the electron contribution to the transport properties when $T_e = T_h$, as well as when $T_e \neq T_h$.

In practice evaluation of the transport properties from the foregoing equations requires a knowledge of the pertinent momentum-transfer collision frequencies $\nu_{eh}^{(1)}$ and hence cross-sections $Q_{eh}^{(1)}$ for the constituents of the plasma in question. These cross sections can be obtained from a variety of sources. Representative data are given in Sec. 14 of Chapter II and an extensive compilation is provided by Brown (1967).

The extension of the foregoing analysis to the calculation of the high-frequency electrical conductivity of a partially ionized plasma follows closely the corresponding calculations of Sec. 2. Thus we write [cf. equation (2.24) and (4.4)]

$$\mathbf{f}^1 = \mathrm{Re}\left[-\frac{e}{kT_e} f_M C(\tilde{D}_1 \mathbf{E}_\parallel^1 + \tilde{D}_2 \mathbf{E}_\perp^1 + \tilde{D}_3 \mathbf{b} \times \mathbf{E}^1) e^{-i\omega t}\right] \tag{4.35}$$

for the part of $\mathbf{f}^1$ that results from a high-frequency field

$$\mathbf{E}^1 \,\mathrm{Re}\, e^{-i\omega t}. \tag{4.36}$$

From equation (4.1), with the term $-\partial \mathbf{f}^1/\partial t$ restored to the left-hand side, it follows that the complex coefficients $\tilde{D}_1$, $\tilde{D}_2$, and $\tilde{D}_3$ must satisfy the equations [cf. equations (4.5)]

$$\begin{aligned} f_M C = &\quad + (-i\omega + \nu_{eH}) f_M C\tilde{D}_1 - \mathbb{C}_{ee}^1[f_M C\tilde{D}_1]/n_e \\ f_M C = &\quad \omega_e f_M C\tilde{D}_3 + (-i\omega + \nu_{eH}) f_M C\tilde{D}_2 - \mathbb{C}_{ee}^1[f_M C\tilde{D}_2]/n_e \end{aligned} \tag{4.37}$$

and

$$0 = -\omega_e f_M C\tilde{D}_2 + (-i\omega + \nu_{eH}) f_M C\tilde{D}_3 - \mathbb{C}_{ee}^1[f_M C\tilde{D}_3]/n_e.$$

With a certain amount of complex algebra (see Exercise 4.5), it is possible to show from equations (4.37) that the conclusions of Sec. 2 with regard to the high-frequency conductivity remain valid. That is, as a result of the high-frequency field (4.36), there is added to the steady current density (4.11) a current density given by

$$\mathbf{J}_e = \mathrm{Re}[(\tilde{\sigma}_\parallel \mathbf{E}_\parallel^1 + \tilde{\sigma}_\perp \mathbf{E}_\perp^1 + \tilde{\sigma}_H \mathbf{b} \times \mathbf{E}^1) e^{-i\omega t}], \tag{4.38}$$

where the complex conductivities $\tilde{\sigma}$ satisfy the relations (2.27); that is,

$$\begin{aligned} \tilde{\sigma}_\parallel &= \sigma_\perp(\omega) + i\sigma_H(\omega), \\ \tilde{\sigma}_\perp &= \tfrac{1}{2}\{\sigma_\perp(\omega + \omega_e) + \sigma_\perp(\omega - \omega_e) + i[\sigma_H(\omega + \omega_e) + \sigma_H(\omega - \omega_e)]\}, \\ \tilde{\sigma}_H &= \tfrac{1}{2}\{\sigma_H(\omega + \omega_e) - \sigma_H(\omega - \omega_e) - i[\sigma_\perp(\omega + \omega_e) - \sigma_\perp(\omega - \omega_e)]\}. \end{aligned} \tag{4.39}$$

and

For a partially ionized plasma the conductivities $\sigma_\perp$ and $\sigma_H$ in equations (4.39) are now to be evaluated by use of the methods of this section.

In the Chapman-Enskog calculation of the transport properties of a mixture of neutral gases it is found that in most cases the convergence of the Sonine polynomial expansions is rapid. Usually only a few terms in the series which correspond to our equations (4.20) and (4.21) are required to obtain accurate values of the various transport properties. The convergence of these expansions for partially ionized plasmas may be significantly slower, however, primarily as a result of the behavior of certain cross sections for electron-neutral collisions. Generally speaking, for plasmas the convergence is less favorable when the electron-neutral cross section in the vicinity of the electron thermal speed increases rapidly with increasing electron speed and when the degree of ionization is low.

We consider first the convergence of the successive approximations for the electron transport properties in the Lorentzian limit of a weakly ionized plasma, for which the exact values of the transport properties are known from the results of Sec. 2. In Fig. 4 the convergence of the electrical and thermal conductivities $\sigma_\parallel^{(\xi)}$ and $\lambda_\parallel^{(\xi)}$ is shown for several power-law models of the electron/heavy-particle collision frequency, for weakly ionized argon at $T_e = 4000°\text{K}$, and for weakly ionized cesium at $T_e = 1500°\text{K}$. For the

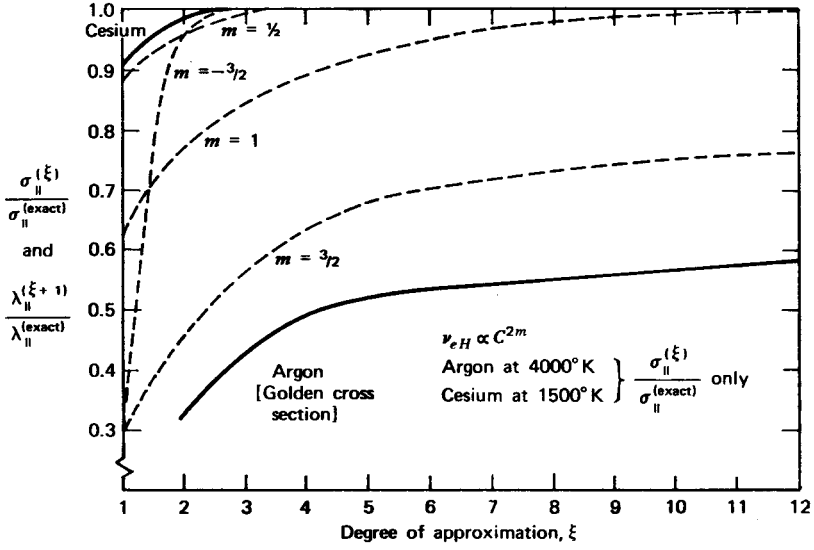



**Figure 4.** Convergence in the Lorentzian limit of the successive Sonine-polynomial values $\sigma_\parallel^{(\xi)}$ and $\lambda_\parallel^{(\xi)}$ as compared to the exact values. To emphasize the trends, the discrete points $\xi = 1, 2, 3, \ldots$ are connected by curves.

power-law models the speed dependence of the momentum-transfer cross section $Q_{eh}^{(1)}(C)$ is represented in terms of a number $m$ by the proportionality

$$Q_{eh}^{(1)}(C) \propto C^{2m-1}.$$

For a single heavy-particle species the collision frequency $\nu_{eH}$ is then of the form

$$\nu_{eH} = n_h C Q_{eh}^{(1)} \propto C^{2m}. \tag{4.40}$$

The values of $\sigma_{\parallel}^{(\xi)}$ and $\lambda_{\parallel}^{(\xi)}$ used to obtain Fig. 4 have been calculated from equations (4.22), (4.23), and (4.25) through (4.28) with $a_{rs}^{(e)}$ and $\omega_e$ set equal to zero. The exact values in the Lorentzian limit are given by equation (2.14a), with $f^0 = f_M$, for $\sigma_{\parallel}$ and the use of the results of Sec. 2 in the formula $\lambda_{\parallel} = \hat{\lambda}_{\parallel} - T_e \phi_{\parallel}(5k/2e + \phi_{\parallel}/\sigma_{\parallel})$ (see Exercise 4.2). For the model cross sections it is found that $\sigma_{\parallel}^{(\xi)}/\sigma_{\parallel}^{(\text{exact})} = \lambda_{\parallel}^{(\xi+1)}/\lambda_{\parallel}^{(\text{exact})}$; for the argon and cesium this does not hold and only $\sigma_{\parallel}^{(\xi)}/\sigma_{\parallel}^{(\text{exact})}$ is shown. It can be seen that for increasing values of $m$ and for argon the convergence is less rapid. We note that $m = -\frac{3}{2}$ corresponds to Coulomb collisions and that $m = \frac{1}{2}$ represents a constant, "hard-sphere" cross section which is independent of $C$. In the case of argon, the pronounced Ramsauer minimum in the cross section at low electron energies has an adverse effect similar to a uniformly increasing cross section (see Chapter II, Fig. 25 which gives the argon cross section measured by Golden used to prepare Figs. 4 and 5).

Of the usual constituents of partially ionized plasmas, argon can be regarded as the least favorable with regard to the convergence of the successive approximations for the transport properties. In fact, Fig. 4 shows that even the twelfth approximation does not give an accurate value for $\sigma_{\parallel}$ of argon in the Lorentzian limit. On the other hand, the convergence is much more rapid in the case of cesium, for which the cross section—shown in Chapter II, Fig. 22—is *relatively* insensitive to electron energy at low energies.

Consistent with Fig. 4, the principal observations which we wish to make with regard to convergence are as follows: In certain cases, such as weakly ionized argon, approximations of a high order are required for accurate calculations. In most cases, however, for most plasma constituents, as for the weakly ionized cesium, the second or third approximation is satisfactory for practical calculations. Furthermore, the error incurred by the use of an approximation of low order is usually no more than approximately a factor of two.

Since as shown in Fig. 4 convergence is good for $m = -\frac{3}{2}$, which corresponds to Coulomb collisions, it would be expected that the convergence would become more rapid as the degree of ionization is increased. This is

confirmed by Fig. 5 which shows the successive approximations for $\sigma_{\parallel}^{(5)}$, $\lambda_{\parallel}^{(5)}$, and $\phi_{\parallel}^{(5)}$ as a function of temperature for atmospheric-pressure equilibrium argon for which $T_e = T_h$. On the abscissa the degree of ionization, based on the use of the Saha equation (II 10.5b), is expressed as $\bar{\nu}_{ei}/\bar{\nu}_{eH}$ [see equation (3.15)]. Here the values of the transport properties given by the twelfth approximation can be regarded as exact, except for the lowest degrees of ionization (cf. Fig. 4).

The convergence of the successive approximations for the thermal conductivity $\lambda_{\parallel}$ and the thermal diffusion coefficient $\phi_{\parallel}$ is similar to that for $\sigma_{\parallel}$. The change in sign of $\phi_{\parallel}$ as the degree of ionization increases is typical of the behavior of real plasmas. On the right in Fig. 5 the values of the transport coefficients for fully ionized plasmas are indicated. For example, for a fully ionized plasma $\sigma_{\parallel}^{(2)} = .978\sigma_{\parallel}^{(\text{exact})}$. (The calculation of the exact value is discussed in the following section.)

The convergence of the perpendicular and Hall coefficients of electrical conductivity is illustrated in Fig. 6. Again it is convenient to consider the Lorentzian limit of Sec. 2 where the exact values can be easily calculated. In this figure $\sigma_{\perp}^{(5)}/\sigma_{\perp}^{(\text{exact})}$ and $\sigma_{H}^{(5)}/\sigma_{H}^{(\text{exact})}$ are shown as functions of magnetic field expressed in terms of $\beta_e = \omega_e/\bar{\nu}_{eH}$, for the case of constant cross sections for which $\nu_{eH} \propto C$. Since $\sigma_{\perp}(B = 0) = \sigma_{\parallel}$, the convergence of $\sigma_{\perp}$ for small $\beta_e$ is the same as that given in Fig. 4 for $m = \frac{1}{2}$. In contrast to results of Figs. 4 and 5 for $\sigma_{\parallel}$, the convergence at some magnetic fields is not monotonic. (For the present example this is apparent from Fig. 6 only for $\sigma_{\perp}$, although for other cross sections a clearly nonmonotonic convergence at some fields is observed for $\sigma_H$ as well.) We note here for later reference that the curves for $\sigma_{\perp}^{(1)}$ and $\sigma_{H}^{(1)}$ show the accuracy of the mean-free-path formulas for the perpendicular and Hall conductivities, since it is readily demonstrated that

$$\sigma_{\perp}^{(1)} = \frac{n_e e^2/m_e \bar{\nu}_{eH}}{1 + \beta_e^2}$$

and (4.41)

$$\sigma_{H}^{(1)} = \frac{(n_e e^2/m_e \bar{\nu}_{eH})\beta_e}{1 + \beta_e^2},$$

where $\beta_e = \omega_e/\bar{\nu}_{eH}$. These results apply for an arbitrary degree of ionization (see Exercise 4.3).

The convergence of $\sigma_{\perp}$ and $\sigma_H$ in the alternate limit of a fully ionized plasma is shown in Fig. 7 for the case of singly charged ions. Here it is appropriate to base the normalizations on $\beta_e = \omega_e/\bar{\nu}_{ei}$ and the exact value of $\sigma_{\parallel}$ for a fully ionized plasma, which is given in the following section.

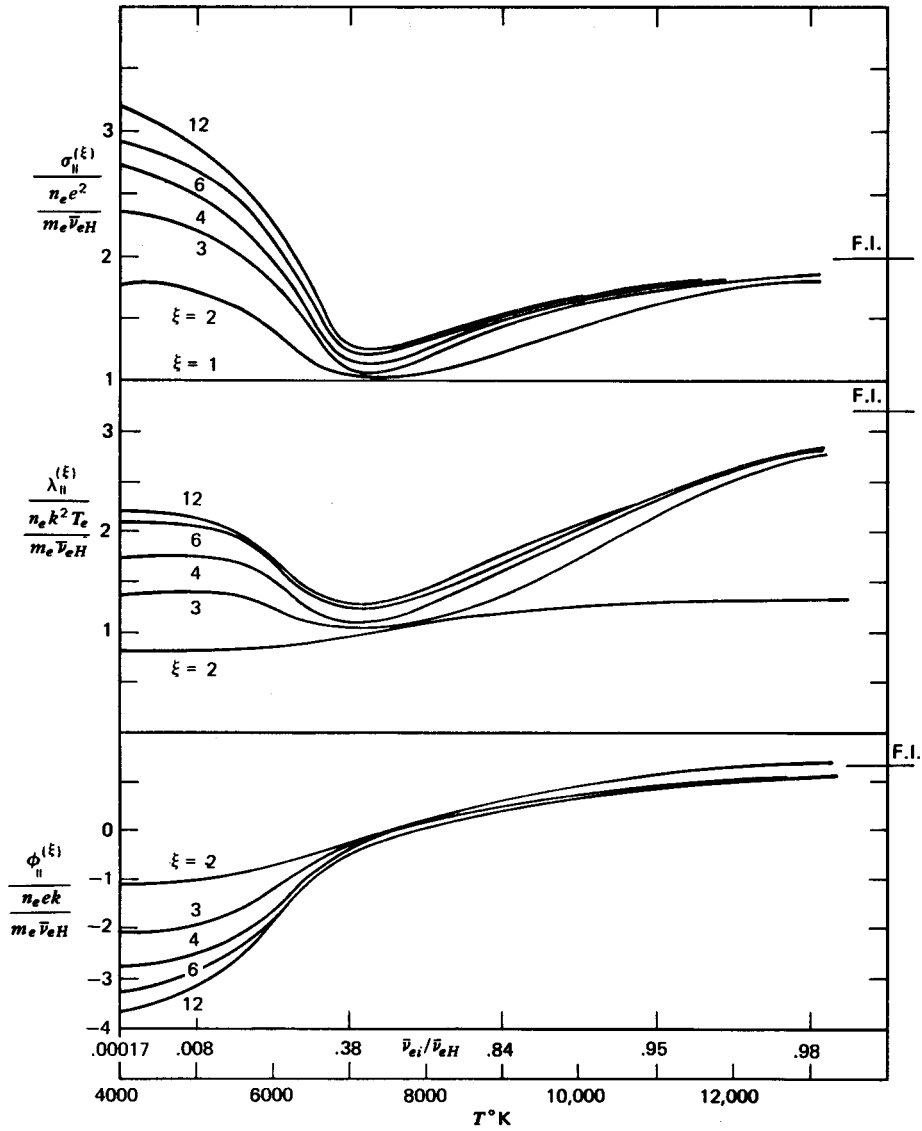



**Figure 5.** Convergence of the successive approximations for $\sigma_{\parallel}$, $\lambda_{\parallel}$, and $\phi_{\parallel}$ for atmospheric-pressure argon.

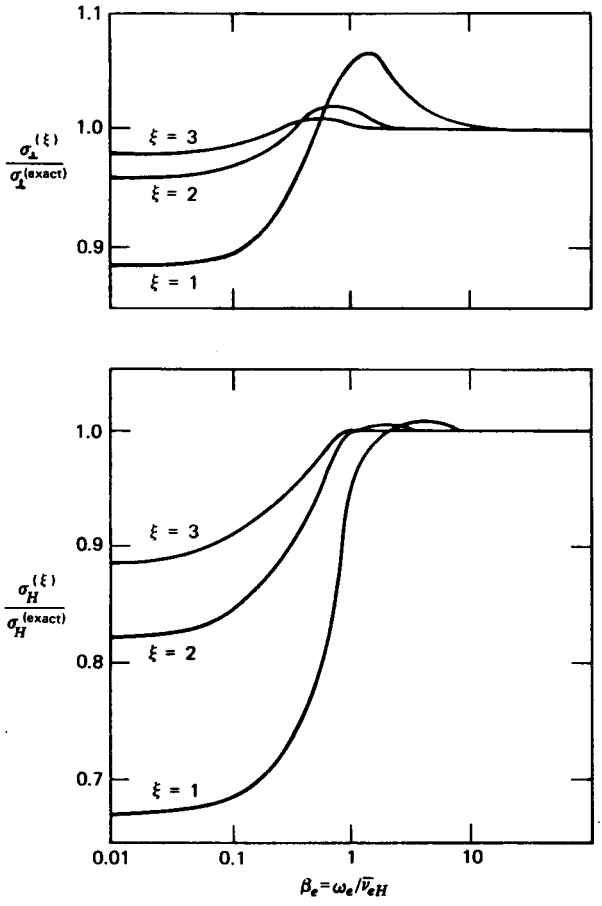



**Figure 6.** The convergence of $\sigma_\perp$ and $\sigma_H$ in the Lorentzian limit as a function of magnetic field for the case of constant cross sections for which $\nu_{eH} \propto C$.

The results given in Figs. 6 and 7 can be taken as reasonably representative of the convergence of the perpendicular and Hall components of the various transport properties in the presence of a magnetic field, with the exception of the somewhat complicated sign behavior of the thermal diffusion coefficient which we have noted in Sec. 2 (see also Fig. 9 in the following section). The relatively slow convergence shown in Fig. 4 for cross sections that are increasing functions of electron energy applies to the perpendicular and Hall components as well, for low values of $\beta_e$. For higher values of $\beta_e$ the convergence is more rapid, in accord with the results of Figs. 6 and 7.

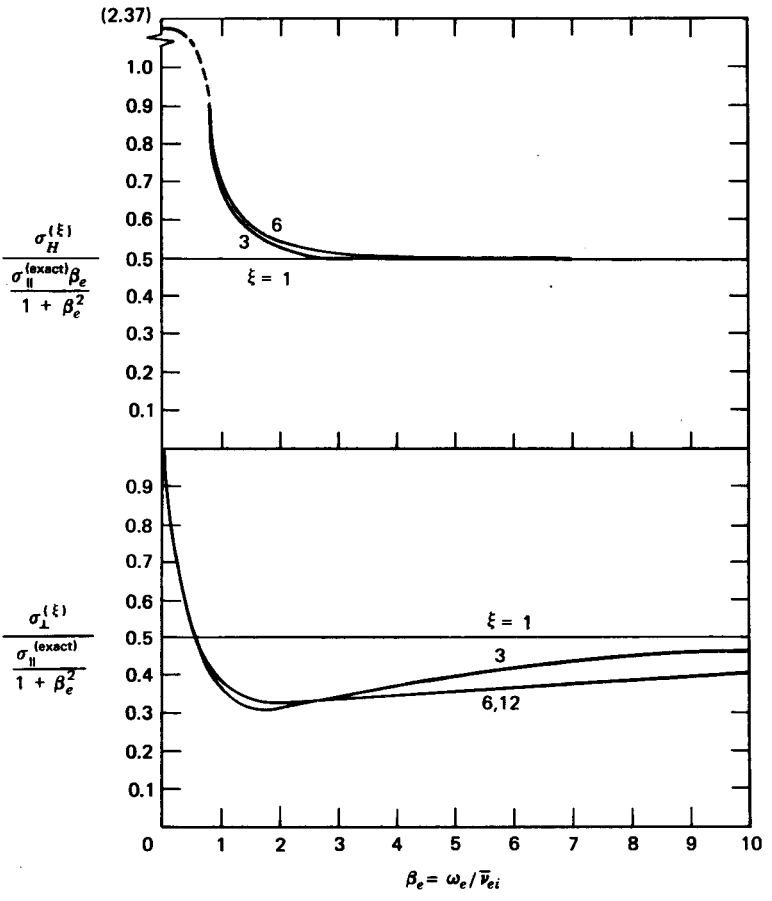



**Figure 7.** Convergence of $\sigma_\perp$ and $\sigma_H$ as a function of magnetic field for a fully ionized plasma.

**Exercise 4.1.** Write the coefficients in the expression for the heat flux $\mathbf{q}_e$ in terms of $\nabla T_e$ and $\mathscr{E}$ as functions of the parallel, perpendicular, and Hall components of $\sigma$, $\lambda'$, and $\phi$.

**Exercise 4.2.** 1. Derive the formula $\lambda_\parallel = \hat{\lambda}_\parallel - T_e \phi_\parallel (5k/2e + \phi_\parallel/\sigma_\parallel)$.
2. Obtain an integral expression for the Lorentzian value of $\lambda'_\parallel$ [cf. equation (7.5a), which follows].

**Exercise 4.3.** Find expressions for the first nonzero approximations for the parallel, perpendicular, and Hall components of $\sigma$, $\lambda'$, $\lambda$, and $\phi$. Express

$\sigma_{\|}^{(1)}$, $\sigma_{\perp}^{(1)}$, and $\sigma_{H}^{(1)}$ in terms of $\bar{\nu}_{eH}$, and compare your results with the mean-free-path formulas (4.41). Compare your expression for $\lambda_{\|}$ with the mean-free-path formula (II 12.19).

**Exercise 4.4.** Show, by means of estimates of the magnitude of the electrical conductivity and thermal-diffusion coefficients, that the thermal-diffusion current produced by a temperature gradient $\nabla kT_e/e$, measured in eV/cm, is similar to that produced by a comparable electric field, measured in V/cm (1 eV $\simeq$ 11,600°K). For simplicity, take $B = 0$. Similarly, estimate the conditions for which terms *other* than those that involve $\nabla T_e$ will be important in equations (4.16) for the heat flux.

**Exercise 4.5.** Demonstrate the validity of equations (4.39) for the complex conductivities $\tilde{\sigma}_{\|}$, $\tilde{\sigma}_{\perp}$, and $\tilde{\sigma}_{H}$.

## 5. FULLY IONIZED PLASMAS

A fully ionized plasma is described as a special case by the equations of the previous section. Since for a fully ionized plasma the number density of neutral particles vanishes, the heavy-particle collision frequency $\nu_{eH}$ becomes $\sum_i \nu_{ei} = \sum_i n_i C Q_{ei}^{(1)}$, where $Q_{ei}^{(1)}$ is given by equation (VII 4.10). Here the summation reflects the fact that, in general, there can be more than one species of ions. With this substitution, the anisotropic part $\mathbf{f}^1$ of the electron distribution function is governed by equations (4.4), (4.8), and (4.9) as before.

In the case where the magnetic field is zero the integro-differential equations (4.8) and (4.9) have been solved for fully ionized plasmas by Spitzer and Härm (1953) by means of numerical integration. For singly charged ions their results yield the following values for the electron transport coefficients:

$$\sigma_{\parallel} = .582 \times \frac{32\sqrt{\pi}\,\epsilon_0^2 (2kT_e)^{3/2}}{e^2\sqrt{m_e}\,\ln\Lambda} = 1.975\,\frac{n_e e^2}{m_e \bar{\nu}_{ei}} \tag{5.1a}$$

$$\lambda^{\parallel} = .236 \times \frac{64\sqrt{\pi}\,\epsilon_0^2 k (2kT)^{5/2}}{e^4\sqrt{m_e}\,\ln\Lambda} = 3.203\,\frac{n_e k^2 T_e}{m_e \bar{\nu}_{ei}} \tag{5.1b}$$

$$\phi_{\parallel} = .273 \times \frac{48\sqrt{\pi}\,\epsilon_0^2 k (2kT_e)^{3/2}}{e^3\sqrt{m_e}\,\ln\Lambda} = 1.389\,\frac{n_e ek}{m_e \bar{\nu}_{ei}} \tag{5.1c}$$

and

$$\psi_{\parallel} = -3.203\,\frac{kT_e}{e}, \text{ where } \bar{\nu}_{ei} = n_i\,\frac{4\sqrt{2\pi}}{3}\left(\frac{m_e}{kT_e}\right)^{3/2}\left(\frac{e^2}{4\pi\epsilon_0 m_e}\right)^2 \ln\Lambda. \tag{5.1d}$$

Numerically, if $T_e$ is expressed in °K the electrical conductivity is

$$\sigma_{\parallel} = 1.53 \times 10^{-2} \frac{T_e^{3/2}}{\ln \Lambda} \text{ mho/m}, \tag{5.2}$$

which was quoted in Chapter II, Sec. 13. Values of the transport coefficients for multiplely charged ions are given by Spitzer (1962).

The analysis of Spitzer and Härm was based on the Fokker-Planck electron-electron collision term, as is the formulation of the previous section. The same results have been obtained by means of the Chapman-Enskog expansion [Chapman and Cowling (1952)] with the use of the Boltzmann collision integral and the Sonine-polynomial expansion by Landshoff (1951), Marshall (1960), and others. The analyses of Spitzer and Härm and Landshoff, and the Sonine-polynomial expansion of the previous section based on the Fokker-Planck collision term, are equivalent, since in each case a Debye cutoff of the Coulomb potential is assumed. Fig. 8 gives corrections to $\sigma_{\parallel}$, $\lambda_{\parallel}$, $\lambda'_{\parallel}$, and $\phi_{\parallel}$ for smaller ln $\Lambda$, calculated on the basis of both the unified theory of Kihara and Aono (1963), and (1971) and the approximate shielded potential (II 4,6) used by Liboff (1959). On the figure the superscripts SH denote the Spitzer-Härm values given by equations (5.1). These corrections have been calculated by Daybelge (1968) from the third Sonine-polynomial approximation [see also Daybelge (1970)]. It can be seen that the less rigorous shielded potential yields results that compare well with those of the unified theory.

The rapid convergence of the Sonine-polynomial expansion in the case of a fully ionized plasma has already been indicated in Figs. 5 and 7 of the previous section for singly charged ions. It is of interest also to compare the exact results with those given by the *ad hoc* use of $\nu_{eH} = \nu_{ei}$ in the Lorentzian formulas of Sec. 2. Numerical values of the various approximations are given in Table 1. The Lorentzian value of $\lambda_{\parallel}$ is again based on the results of Sec. 2 and the formula

$$\lambda_{\parallel} = \hat{\lambda}_{\parallel} - T_e \phi_{\parallel} \left( \frac{5k}{2e} + \phi_{\parallel} / \sigma_{\parallel} \right)$$

(see Exercise 4.2).

In the presence of a magnetic field the Sonine-polynomial expansion of the previous section can be used to calculate accurate values of the perpendicular and Hall components of the various electron transport coefficients. These components of the electrical conductivity were shown in Fig. 7 of that section. To complete the description, the twelfth approximation for the thermal conductivity and thermal diffusion coefficients is given

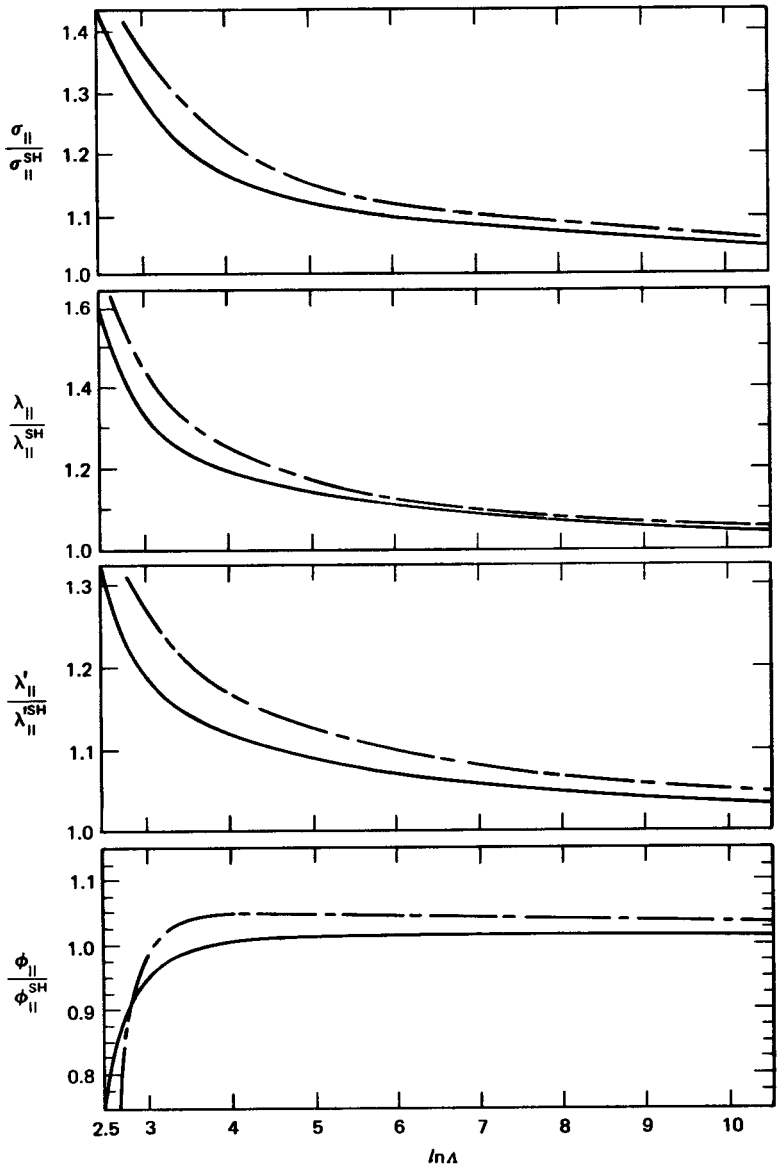



**Figure 8.** Corrections to the electron transport coefficients of a fully ionized plasma, based on the unified theory of Kihara and Aono (——) and on a shielded Coulomb potential (— — —).

**Table 1 Comparison of the Sonine-polynomial and Lorentzian Approximations for the Transport Coefficients of a Fully Ionized Plasma with the Exact Values Calculated Numerically by Spitzer and Härm (1953) for Singly Charged Ions.**

| | $\frac{\sigma_\parallel}{\sigma_\parallel^{(\text{exact})}}$ | $\frac{\lambda_\parallel}{\lambda_\parallel^{(\text{exact})}}$ | $\frac{\phi_\parallel}{\phi_\parallel^{(\text{exact})}}$ |
|---|---|---|---|
| $\xi = 1$ | .506 | — | — |
| $\xi = 2$ | .978 | 0.418 | 1.12 |
| $\xi = 3$ | .987 | 0.9875 | 1.00 |
| Lorentzian | 1.72 | 4.24 | 3.67 |

here in Fig. 9, again for the case of singly charged ions. We note that for the perpendicular thermal-diffusion coefficient the use of the factor $1 + \beta_e^2$ in the normalization loses much of its significance, since $\phi_\perp$ changes sign as the magnetic field increases.

In addition to the foregoing electron transport properties, it is pertinent to consider the transport associated with the ions. In particular, the viscous stress results almost entirely from momentum transport by the ions and hence can be related to the ion viscosity coefficient [see equation (6.1)]

$$\eta_i = \frac{0.406(4\pi\epsilon_0)^2\sqrt{m_i}(kT_i)^{5/2}}{Z^4e^4 \ln \Lambda}, \tag{5.3}$$

where $Z$ is the ion charge number. This result, based on calculations discussed by Shkarofsky, Johnston, and Brachynski (1966), Sec. 8.8, applies in the usual situation where the ratio of the ion cyclotron frequency to the ion collision frequency is small.

As a result of the condition

$$\sum_i \rho_i \mathbf{U}_i + \rho_e \mathbf{U}_e = 0, \tag{5.4}$$

which follows from the definition (VII 2.9b) of the diffusion velocities relative to the mass velocity $\mathbf{u}$, the diffusion velocities of the ions are always small in a fully ionized gas and need not be considered. Thus, whether or not a magnetic field is present, the contribution of the ions to any component of the conduction current density is negligible. In the absence of a magnetic field, the electron thermal conductivity and thermal diffusion coefficients are much larger than those of the ions and hence the ion heat flux can also

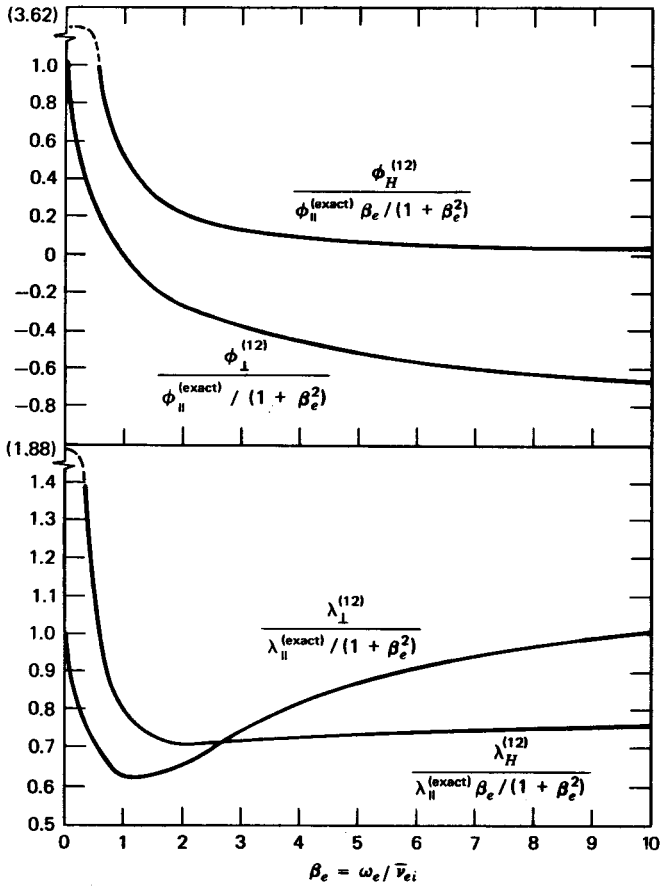



**Figure 9.** The perpendicular and Hall components of thermal conductivity and thermal diffusion for a fully ionized plasma.

be neglected. The electron heat flux perpendicular to a magnetic field, however, will be small if $\omega_e/\bar{\nu}_{ei}$ is large, as can be seen from Fig. 9. In this case the perpendicular ion heat flux must be considered. The appropriate transport coefficients have been calculated by several authors and are discussed by Shkarofsky, Johnston, and Brachynski (1966), Sec. 8.7. [We note that an error in the perpendicular thermal conductivity calculated by Marshall (1960) has been corrected by Vaughnan-Williams and Haas (1961).]

The role of heavy-particle transport effects in the more general situation of a partially ionized plasma is discussed in the following section.

**Exercise 5.1.** Show by the use of suitable mean-free-path arguments that for a fully ionized plasma with $B = 0$ it is reasonable to expect that the electron contribution to the viscosity and the ion contribution to the thermal conductivity can be neglected. Compare your estimates of $\eta_i$ and $\lambda_{||}$ with the exact expressions.

**Exercise 5.2.** Obtain approximate expressions for $\sigma_{||}$ and $\lambda_{||}$ by the *ad hoc* application of the Lorentzian analysis of Sec. 2 to a fully ionized plasma. Your results should agree with the values given in Table 1.

**Exercise 5.3.** For a fully ionized plasma evaluate $\sigma_{||}^{(1)}$.

## 6. HEAVY-PARTICLE TRANSPORT PROPERTIES

In the previous sections, we have discussed almost exclusively the calculation of electron transport properties. To obtain a complete description of transport phenomena in a collision-dominated plasma, however, it is necessary to consider also the heavy-particle transport properties. Thus, for example, the viscous stress depends on the heavy-particle viscosity coefficient and the total heat flux in a partially ionized plasma depends on the heavy-particle, as well as the electron, thermal condictivity.

In the absence of a magnetic field, a rigorous means of calculation of the heavy-particle transport properties—based on the classical Chapman-Enskog theory for gas mixtures—is provided by Hirschefelder, Curtiss, and Bird (1954), Chapter 7. There it is shown that the *total* viscous stress $\tau_{\alpha\beta}$ and heat flux $\mathbf{q}$ and the species diffusion velocities $\mathbf{U}_s$ are related to gradients and forces by the following expressions:

$$\tau_{\alpha\beta} = \eta\left[\frac{\partial u_\alpha}{\partial x_\beta} + \frac{\partial u_\beta}{\partial x_\alpha} - \frac{2}{3}(\nabla \cdot \mathbf{u})\delta_{\alpha\beta}\right], \tag{6.1}$$

$$\mathbf{q} = \frac{5}{2}kT\sum_s n_s \mathbf{U}_s - \lambda'\nabla T - nkT\sum_s \frac{D_s^T}{\rho_s}\mathbf{d}_s, \tag{6.2}$$

and

$$\mathbf{U}_s = \frac{n^2}{n_s\rho}\sum_r m_r D_{sr}\mathbf{d}_r - \frac{1}{\rho_s}D_s^T \nabla \ln T, \tag{6.3}$$

The effective driving forces $\mathbf{d}_s$ for diffusion are defined by

$$\mathbf{d}_s \equiv \nabla\left(\frac{n_s}{n}\right) + \left(\frac{n_s}{n} - \frac{\rho_s}{\rho}\right)\nabla \ln p \tag{6.4}$$

$$- \frac{\rho_s}{p\rho}\left(\frac{\rho}{m_s}\mathbf{F}_s - \sum_r n_r \mathbf{F}_r\right).$$

Here $\eta$ is the overall viscosity coefficient, $\lambda'$ is an overall thermal-conductivity coefficient, and $D_{sr}$ and $D_s^T$ are species diffusion and thermal-diffusion coefficients, respectively. The other symbols in equations (6.1) through (6.4) are defined in Chapter VII, Secs. 2 and 3. As in Sec. 4, the thermal-conductivity coefficient $\lambda'$ is not the usual coefficient $\lambda$, which relates the heat flux to the temperature gradient when the diffusion velocities are zero. We note that, since the classical Chapman-Enskog theory applies to quasiequilibrium situations where the temperatures of the various species are the same, only a single temperature appears in these equations.

Hirschfelder, Curtiss, and Bird show how the various transport coefficients in equations (6.1) through (6.4) can be evaluated, with the use of Sonine-polynomial expansions, in terms of the solutions of linear algebraic equations. As in Sec. 4, these solutions can be expressed in the form of ratios of determinants of collision integrals. The order of the determinants involved is equal to the product of the number of species and the order of the Sonine-polynomial approximation; hence, the formulation is complicated at best. To directly apply this theory to a plasma, the electrons and ions must be regarded as on an equal footing with the other species in the gas mixture. The theory can be used, however, to calculate separately the various heavy-particle diffusion and thermal-diffusion coefficients and the heavy-particle contributions to the viscosity coefficient and the thermal-conductivity coefficient $\lambda'$. Nevertheless, each of these calculations still involves electron collision integrals.

The direct application of the results of Hirschfelder, Curtiss, and Bird to the calculation of the heavy-particle transport properties of a partially ionized plasma suffers from two defects. First, the theory applies to quasiequilibrium plasmas with a single temperature for all species and, second, the results do not incorporate the simplifications associated with the small mass of the electron. Hence they are more complicated than necessary. Daybelge (1968) has examined this situation from the point of view of Chapter VII, Sec. 5, where the expansion of the collision terms for small electron mass was discussed. He finds that for a collision-dominated plasma the only significant effect of the electrons on the heavy-particle distribution functions results from a collisional transfer of momentum associated with the diffusion of the electrons. The reasons for this are that the electrons do not significantly affect the heavy-particle viscosity coefficients and that the heavy-particle diffusion, thermal-diffusion, and thermal-conductivity coefficients depend on equations analogous to equations (VII 6.40) or (4.1), which can be interpreted as speed-dependent momentum equations for each heavy-particle species. The effect of collisions with electrons can be represented in each of these equations by the rate per heavy particle of momentum transfer from the electrons to the heavy-particle species in question.

The rate of momentum transfer from the electrons to the heavy particles can be satisfactorily calculated from a consideration of only the zero-order terms in $m_e/m_h$ in the expansions of Chapter VII, Sec. 5. Thus, since in this approximation the heavy particles can be regarded as stationary, the differential rate per heavy particle of momentum transfer, integrated over the azimuthal angle $\phi$, is

$$n_e m_e \mathbf{C}(1 - \cos\chi) f_e(\mathbf{C}) C I_{eh}(C, \chi) 2\pi \sin\chi \, d\chi \, d^3C.$$

Integration of this expression over all deflection angles and electron velocities yields for the required rate of collisional momentum transfer

$$n_e m_e \int_{-\infty}^{\infty} \mathbf{C} C Q_{eh}^{(1)}(C) f_e(\mathbf{C}) \, d^3C \tag{6.5}$$

$$= n_e m_e \overline{(\mathbf{C} C Q_{eh}^{(1)})}_e,$$

where $Q_{eh}^{(1)}$ is the momentum-transfer cross section defined by equation (VII 4.9). The notable feature of this result (6.5) is that to this approximation the rate of momentum transfer is independent of the heavy-particle velocity and therefore has the same effect as an external force. Accordingly, Daybelge finds that for purposes of calculation of the heavy-particle transport properties, the electron collision terms in the heavy-particle Boltzmann equations can be omitted, if in their place is incorporated on the left-hand side of each Boltzmann equation an effective force given by equation (6.5). We note that the expression (6.5), when multiplied by $n_H$, is just equal to the integrated rate of momentum *loss* to the electrons that would be found from the multiplication of the term $v_{eH}\mathbf{f}^1$ in equation (4.1) by $n_e m_e C (4/3)\pi C^2 \, dC$ and integration over $C$ [see the second of equations (VII 6.19)].

As a result of the foregoing considerations, Daybelge gives the following simplified prescription for the evaluation of heavy-particle transport effects in partially ionized plasmas: To within an error of the order of $\sqrt{m_e/m_h}$, equations (6.1) through (6.4) can be used to evaluate the total viscous stress, the *heavy-particle* heat flux—which is to be added to the electron heat flux (4.12) or (4.14)—and the heavy-particle diffusion velocities. The transport properties that appear in these expressions are to be evaluated according to the methods of Hirschfelder, Curtiss, and Bird. Their formulas, however, are to be applied to a hypothetical gas mixture of only neutral particles and ions at a common temperature $T$ which may be different from the electron temperature $T_e$. The electrons are to be ignored entirely in the calculation of the heavy-particle transport properties and omitted from the summations in equations (6.2) and (6.3). In equation (6.4) for the driving force $d_s$, the effective force (6.5) is to be added to the external force $\mathbf{F}_s$ and the electrons are to be included in the evaluation of $n$, $p$, and the final summation. As a consequence of the foregoing, it is possible to evaluate

the heavy-particle transport properties on the basis of low-order Sonine-polynomial approximations—which, as discussed by Hirschfelder, Curtiss, and Bird, are usually sufficiently accurate—and at the same time use the methods of Sec. 4 to obtain when necessary higher-order approximations for the electron transport properties.

The experienced reader may raise the objection to this prescription that for the hypothetical mixture of neutrals and ions the condition $\Sigma_n \rho_n \mathbf{U}_n + \rho_i \mathbf{U}_i = 0$ would apparently be imposed by use of the results of Hirschfelder, Curtiss, and Bird, whereas for the actual plasma the condition on the diffusion velocities must be $\sum_n \rho_n \mathbf{U}_n + \rho_i \mathbf{U}_i + \rho_e \mathbf{U}_e = 0$ [see equation (VII 2.9c)]. Daybelge (1968) shows, however, that this point does not actually present any difficulty and that to order $\sqrt{m_e/m_h}$ no error is thereby introduced in the calculation of the heavy-particle transport properties.

To carry out these simplifications in detail it is necessary to evaluate the effective forces (6.5) on the heavy particles. Since the analysis of Sec. 4 for the electron distribution function is uncoupled from the calculation of the heavy-particle distribution functions, these effective forces can be found independently, from the relation [see equations (VII 6.19)]

$$n_e m_e \overline{(\mathbf{C} C Q_{eh}^{(1)})}_e = n_e m_e \frac{4\pi}{3} \int_0^\infty C^4 Q_{eh}^{(1)} \mathbf{f}^1 \, dC. \tag{6.6}$$

With the use of equation (4.4) and the subsequent results of Sec. 4, $\mathbf{f}^1$ can be expressed in terms of the Sonine-polynomial expansion coefficients, which in turn can be found from the solution of equations (4.25) and (4.26). Evaluation of the integral on the right-hand side of equation (6.6), as described in detail by Daybelge, is thus reasonably straightforward. If, however, the heavy-particle diffusion velocities are not small when compared with the electron diffusion velocity this procedure must be re-examined. In this case, equation (6.6) no longer accurately gives the rate of momentum transfer from the electrons to the heavy particles, as is reflected by the term $-n_e \nu_{eh}^{(1)} \mathbf{U}_h \, \partial f^0/\partial C$ in the electron equation (VII 6.40). With respect to the calculation of the heavy-particle transport properties, it turns out that this does not matter, since when $U_h \gg (m_e/m_h)^{1/2} U_e$ the momentum transfer in question will be small and the term (6.6) can be omitted from the heavy-particle equations. On the other hand, in this case to calculate the electron transport properties the theory of Sec. 4 must now be extended to account for the effects of the heavy-particle diffusion velocities on the electrons.

In many situations it is not necessary in practice to actually evaluate the integral in equation (6.6). This term does not effect the calculation of the viscosity or the heavy-particle thermal-conductivity $\lambda'$ but only enters in the evaluation of the heavy-particle diffusion velocities and in the thermal-diffusion term in the heavy-particle heat flux (6.2). This is in accord with the

numerical calculations of DeVoto (1967), who evaluated the $\eta$ and $\lambda$ for partially ionized atmospheric-pressure argon, first using the complete formalism of Hirschfelder, Curtiss, and Bird and then neglecting entirely the presence of the electrons [including the term (6.6)] in the evaluation of $\mu$ and the heavy-particle contribution to $\lambda$. Over a range of temperatures from 5000°K to 20,000°K, he found less than a 1% change in both coefficients, when the same Sonine-polynomial approximation is used for each method.

The foregoing discussion has been limited, nominally, to the case of zero magnetic field. Extension of the Chapman-Enskog theory of a multicomponent plasma to the magnetic-field case has been worked out by DeVoto (1969) and by Bisnovatyy-Kogan (1964) [see Kalikhman (1967)]. Although this theory already incorporates certain features associated with the small mass of the electron, Daybelge (1968) shows that the results can be further simplified in exactly the same way as in the case of zero magnetic field. Except for very strong magnetic fields or low densities, however, the ratio of the ion cyclotron frequency $eB/m_i$ to the ion collision frequency will be small. When this is so, the simplified equations for the heavy-particle transport properties reduce to those of the zero-magnetic-field case. Accordingly, under these conditions our previous discussion of the use of the results of Hirschfelder, Curtiss, and Bird is directly applicable and the magnetic field enters only in the calculations of Sec. 4 for the electron transport effects and, thereby, in the evaluation of the effective forces (6.6).

Having established the sources that can be used for the calculation of the heavy-particle transport properties, we return in the following section to the consideration of electron transport properties. In particular, with the use of results of Sec. 4, we shall examine the accuracy of various approximate formulas for these properties.

## 7. MIXTURE RULES

On the basis of the theory of Sec. 4 the electron transport properties can be calculated to any accuracy desired by means of the Sonine-polynomial expansions. These calculations incorporate the simplifications associated with the small mass of the electron and hence are much less complicated than the corresponding classical calculations for a gas mixture. None the less, in many practical cases it is desirable or necessary to use for the transport properties approximate formulas that can be evaluated with a minimum of numerical effort. In this section we discuss such formulas, called mixture rules, and assess their accuracy in comparison with the exact theory of Sec. 4. Throughout the section we shall restrict our attention to electron transport properties.

We first consider the approximate calculation of the parallel electrical conductivity $\sigma_\parallel$, which applies in the absence of or in the direction parallel to a magnetic field. The simplest expression for $\sigma_\parallel$ is the mean-free-path formula

$$\sigma_\parallel \simeq \frac{n_e e^2}{m_e \bar{\nu}_{eH}}. \tag{7.1}$$

In practice, various methods are used to evaluate the collision frequency in equation (7.1), as discussed in Chapter II, Sec. 6. These differences in the form of the collision frequency can be interpreted in terms of different methods of averaging the speed-dependent cross section $Q_{eh}^{(1)}$. If, as indicated in equation (7.1), the well-defined expression (3.15) is used for the collision frequency, with $\bar{\nu}_{eH} = \sum_h \bar{\nu}_{eh}$, the mean-free-path formula (7.1) becomes identical to the first Sonine-polynomial approximation $\sigma_\parallel^{(1)}$ in the theory of Sec. 4. Hence, Figs. 4 and 5 of Sec. 4, which show the convergence of the Sonine-polynomial approximations, can be used to assess the accuracy of equation (7.1). The mean-free-path formula (7.1) is in error by a factor of about two in the limit of a fully ionized plasma and, for some electron-neutral cross sections, by even larger factors in the limit of a weakly ionized plasma. These inaccuracies can be avoided by the use of the approximate conductivity $\sigma_\parallel^{(\mathrm{add})}$, given by

$$\frac{1}{\sigma_\parallel^{(\mathrm{add})}} = \frac{1}{\sigma_{en}} + \frac{1}{\sigma_{ei}}. \tag{7.2}$$

Equation (7.2) is based on the suggestion of Lin, Resler, and Kantrowitz (1955) that the resistivities associated with the neutrals and ions be added. If the Lorentzian expression (2.14a), with $f^0 = f_M$ and applied only to the neutral species, is used for $\sigma_{en}$ and the Spitzer-Härm expression (5.1a) is used for $\sigma_{ei}$, the conductivity $\sigma_\parallel^{(\mathrm{add})}$ is necessarily correct in the limits of both weakly ionized and fully ionized plasmas. Although the formula (7.2) is accordingly preferable to equation (7.1), $\sigma_\parallel^{(\mathrm{add})}$ may still be in error by as much as a factor of about two at intermediate degrees of ionization, as is shown in Fig. 10.

There exists an alternate mixture rule which yields accurate values for $\sigma_\parallel$ over the complete range of ionization. This is the formula proposed by Frost (1961), in which the Lorentzian conductivity expression (2.14a) is used with a modified electron-ion collision frequency such that the resultant formula reduces for singly charged ions to the exact Spitzer-Härm value (5.1a) in the limit of a fully ionized plasma. The Frost conductivity $\sigma_\parallel^{(F)}$ is given by

$$\sigma_\parallel^{(F)} = \frac{4\pi n_e e^2}{3kT_e} \int_0^\infty \frac{C^4 f_M}{\nu_\sigma^{(F)}}\, dC, \tag{7.3a}$$

where

$$\nu_\sigma^{(F)} = \sum_n \nu_{en}^{(1)} + 0.476 \frac{8\pi n_i}{C^2} \left(\frac{m_e}{2kT_e}\right)^{1/2} \left(\frac{e^2}{4\pi\epsilon_0 m_e}\right)^2 \ln \Lambda. \tag{7.3b}$$

Comparison of the ion contribution to the collision frequency (7.3b) with $\nu_{ei}^{(1)}$ given by equation (VII 6.37) shows that not only is the numerical factor changed, but in addition the $C^{-3}$ speed-dependence of $\nu_{ei}^{(1)}$ is replaced by $C^{-2}(m_e/2kT_e)^{1/2}$. Although this modification of the speed-dependence of $\nu_{ei}^{(1)}$ is based on certain calculations of Hwa (1957), there is no rigorous justification for the Frost formula (7.3). Nevertheless, as shown by Schweitzer and Mitchner (1966) and for partially ionized argon in Fig. 10, the Frost formula is remarkably accurate. In fact, for the lower degrees of ionization shown in Fig. 10 the Frost conductivity, which reduces to the exact value in the Lorentzian limit, is presumably more accurate than even the twelfth Sonine-polynomial approximation. This results from the poor convergence of the Sonine-polynomial expansion in the limit of a weakly ionized plasma for the "worst-case" of argon, which was shown in Fig. 4 of Sec. 4. For higher degrees of ionization, where $\sigma_\parallel^{(F)}$ and $\sigma_\parallel^{(1\,2)}$ agree, the Sonine-polynomial expansion converges more rapidly (cf. Fig. 5) and the values of $\sigma_\parallel^{(1\,2)}$ can be taken as exact.

Mixture rules for the approximate calculation of the thermal-conductivity and thermal-diffusion coefficients $\lambda'_\parallel$, $\lambda_\parallel$, and $\phi_\parallel$ have not been as well developed as in the case of $\sigma_\parallel$. For these properties the use of the lowest possible Sonine-polynomial approximation is more complicated since this now requires $\xi = 2$. The accuracy of this approach is illustrated in Figs. 4 and 5 of Sec. 4. Alternatively, mean-free-path considerations suggest that the thermal conductivity $\lambda_\parallel$ should be given to within a numerical factor by

$$\lambda_\parallel \simeq \frac{n_e k^2 T_e}{m_e \bar{\nu}_{eH}}. \tag{7.4}$$

From the normalization used in Figs. 4 and 10, it can be seen that for argon equation (7.4) gives $\lambda_\parallel$ to within a factor of three. As a result of the change in sign of the thermal-diffusion coefficient with increasing degree of ionization (see Fig. 5), there is no meaningful counterpart for $\phi_\parallel$ to the simple formula (7.4). One can say, however, that the *magnitude* of $\phi_\parallel$ is not much greater than $n_e ek/m_e \bar{\nu}_{eH}$, as can be seen from Fig. 5.

Kruger, Mitchner and Daybelge (1968) have proposed mixture rules for $\lambda'_\parallel$ and $\phi_\parallel$ which are similar to the Frost conductivity (7.3). The formulas for these quantities, denoted by $\lambda_\parallel'^{(F)}$ and $\phi_\parallel^{(F)}$, are

$$\lambda_\parallel'^{(F)} = \frac{4\pi}{3} n_e k \int_0^\infty \frac{1}{\nu_\lambda^{(F)}} \left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right)^2 C^4 f_M \, dC, \tag{7.5a}$$

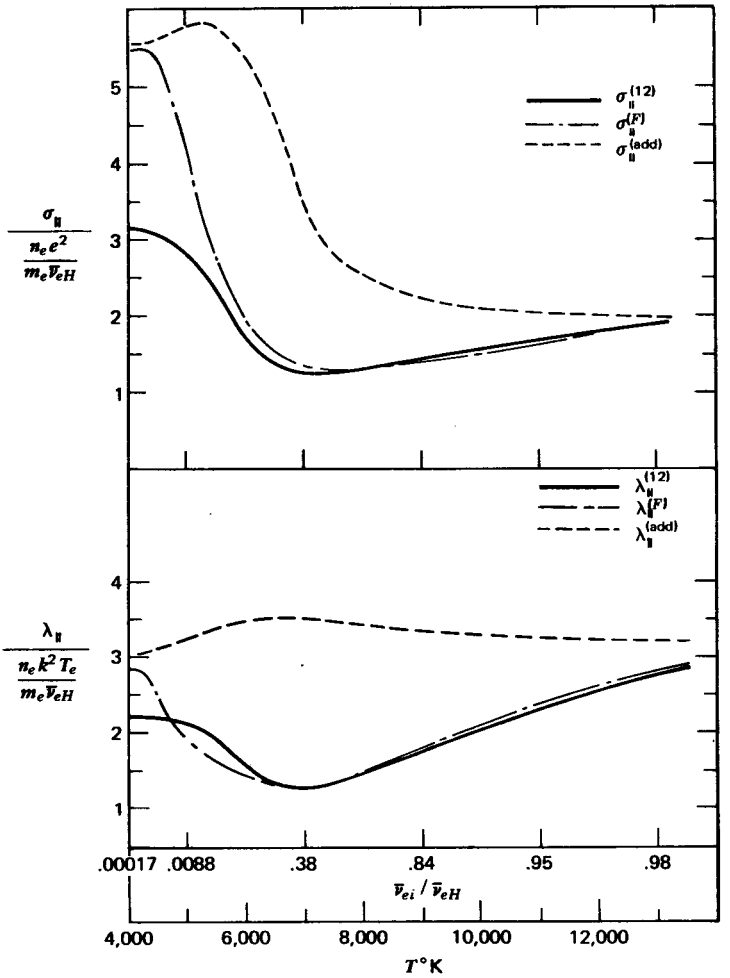



**Figure 10.** Comparison of the mean-free-path and Frost values for $\sigma_{\parallel}$ and $\lambda_{\parallel}$ with the values given by the twelfth approximation for atmospheric-pressure argon.

where

$$\nu_{\lambda'}^{(F)} = \sum_n \nu_{en}^{(1)} + 1.012 \frac{8\pi n_i}{C^2} \left(\frac{m_e}{2kT_e}\right)^{1/2} \left(\frac{e^2}{4\pi\epsilon_0 m_e}\right)^2 \ln \Lambda, \tag{7.5b}$$

and

$$\phi_{\parallel}^{(F)} = \frac{4\pi}{3} \frac{en_e}{T_e} \int_0^\infty \frac{1}{\nu_\phi^{(F)}} \left(\frac{m_e C^2}{2kT_e} - \frac{5}{2}\right) C^4 f_M \, dC. \tag{7.6a}$$

where

$$\nu_\phi^{(F)} = \sum_n \nu_{en}^{(1)} + 0.6776 \frac{8\pi n_i}{C^2} \left(\frac{m_e}{2kT_e}\right)^{1/2} \left(\frac{e^2}{4\pi\epsilon_0 m_e}\right)^2 \ln \Lambda. \tag{7.6b}$$

From these expressions a "Frost" value $\lambda_{\parallel}^{(F)}$ for the usual thermal conductivity can be calculated from the relation [cf. equations (4.15)]

$$\lambda_{\parallel}^{(F)} = \lambda_{\parallel}'^{(F)} - T_e(\phi_{\parallel}^{(F)})^2/\sigma_{\parallel}^{(F)}. \tag{7.7}$$

As with the electrical conductivity, the "Frost" values of the thermal-conductivity and thermal-diffusion coefficients, calculated from equations (7.5) through (7.7), have been found to agree well with the exact values. This is illustrated for the thermal conductivity of argon in Fig. 10. Again for low degrees of ionization the values of $\lambda_{\parallel}^{(F)}$ are presumably more accurate than those given by the twelfth Sonine-polynomial approximation, as a result of the poor convergence of the Sonine-polynomial expansions for weakly ionized argon.

We can conclude from the foregoing that the mixture rules of the Frost type, being both accurate and reasonably easy to use, are to be preferred for the practical calculation of transport properties in the absence of a magnetic field. In fact, once the accuracy of these formulas has been established, it would appear that the considerable additional complexity of the use of high-order Sonine-polynomial approximations is hardly warranted, particularly in view of the inevitable uncertainties in cross sections. Corrections for smaller ln $\Lambda$, as proposed by Daybelge (1968) and (1970), can be made by dividing the second terms in equations (7.3b), (7.5b), and (7.6b) by $\sigma_{\parallel}/\sigma_{\parallel}^{SH}$, $\lambda_{\parallel}'/\lambda_{\parallel}'^{SH}$, and $\phi_{\parallel}/\phi_{\parallel}^{SH}$ from Fig. 8 of Sec. 5.

The situation with regard to mixture rules in the presence of a magnetic field is more complicated and somewhat less satisfactory as a result of the additional parameter, the field strength, and the increased number of transport properties. As before, when only limited accuracy is required either mean-free-path formulas or the lowest Sonine-polynomial approximation can be used. In the mean-free-path formulas the value of the parallel component of the transport property is multiplied by $(1 + \beta_e^2)^{-1}$ and by $\beta_e(1 + \beta_e^2)^{-1}$ to obtain the perpendicular and Hall components, respectively.

Although in these expressions $\beta_e$ can be evaluated in various ways, for definiteness we use the form $\beta_e = \omega_e/\bar{\nu}_{eH}$ based on $\bar{\nu}_{eH} = \sum_h \bar{\nu}_{eh}$ and the definition (3.15) of $\bar{\nu}_{eh}$. Figs. 6, 7, and 9 of Secs. 4 and 5 illustrate the fact that such formulas yield values that are accurate to within a factor of two or three. In the case of the perpendicular thermal diffusion coefficient, however, the complicated change in sign with both increasing degree of ionization and increasing $\beta_e$ makes the mean-free-path formula of limited utility. We note that by equations (4.41) the simple mean-free-path values for the perpendicular and Hall conductivities

$$\sigma_\perp \simeq \frac{n_e e^2/m_e \bar{\nu}_{eH}}{1 + \beta_e^2}$$

and

$$\sigma_H \simeq \frac{(n_e e^2/m_e \bar{\nu}_{eH})\beta_e}{1 + \beta_e^2}$$

are given in Fig. 6 by the curves for $\sigma_\perp^{(1)}$ and $\sigma_H^{(1)}$.

To obtain more accurate values of the electrical conductivity in the presence of a magnetic field at least two additional methods are available. First, to the extent that the electron-neutral collision frequency can be described by a power-law dependence on electron speed— in our notation $\sum_n \nu_{en} \propto C^{2m}$—the results of Shkarofsky, Johnston, and Bachynski (1966), Sec. 8.5, can be used to obtain directly the fourth Sonine-polynomial approximations to $\sigma_\perp$ and $\sigma_H$. Since in most cases the fourth approximation will itself be sufficiently accurate, the primary uncertainty in this method results from the power law representation of the electron-neutral collision frequency. The second method, proposed by Schweitzer and Mitchner (1967), consists of an extension of the Frost mixture rule to $\sigma_\perp$ and $\sigma_H$. This method uses the Lorentzian expressions (2.14b) and (2.14c) for $\sigma_\perp$ and $\sigma_H$, incorporating effective electron-ion collision frequencies which are such that $\sigma_\perp$ and $\sigma_H$ reduce to known values in the limit of a fully ionized plasma. These effective collision frequencies depend on the magnetic-field strength and also on the order of the Sonine-polynomial approximation used to calculate $\sigma_\perp$ and $\sigma_H$ in the fully ionized limit. The curves for these collision frequencies given by Schweitzer and Mitchner (1967) are based in part on the third Sonine-polynomial approximations. Somewhat more accurate values, which incorporate the eleventh approximations, are given by Kruger, Mitchner, and Daybelge (1968), who also give similar formulas for $\lambda'_\perp$, $\lambda'_H$, $\phi'_\perp$, and $\phi'_H$. Comparison of the results obtained by this method with exact values shows good agreement.

With respect to the accuracy of approximate formulas for the electrical conductivity in the presence of a magnetic field, it is instructive also to

examine certain consequences of the usual mean-free-path representation of Ohm's law, namely,

$$\mathbf{J}_e \simeq \sigma_{\parallel} \mathscr{E} - \beta_e \mathbf{J}_e \times \mathbf{b}, \tag{7.8}$$

where $\mathbf{b} = \mathbf{B}/B$. According to equation (7.8), in the plane perpendicular to the magnetic field the angle between $(\mathbf{J}_e)_\perp$ and the generalized electric field $\mathscr{E}_\perp = \mathbf{E}'_\perp + \nabla_\perp p_e/en_e$ is given by $\tan^{-1} \beta_e$. In addition, the current density $(\mathbf{J}_e)_\perp$ is equal to the product of the "scalar" electrical conductivity—in our notation $\sigma_{\parallel}$—and the component of the field $\mathscr{E}_\perp$ in the direction of the current. These approximations are often used in the interpretation of the electrical characteristics of crossed-field devices such as MHD generators

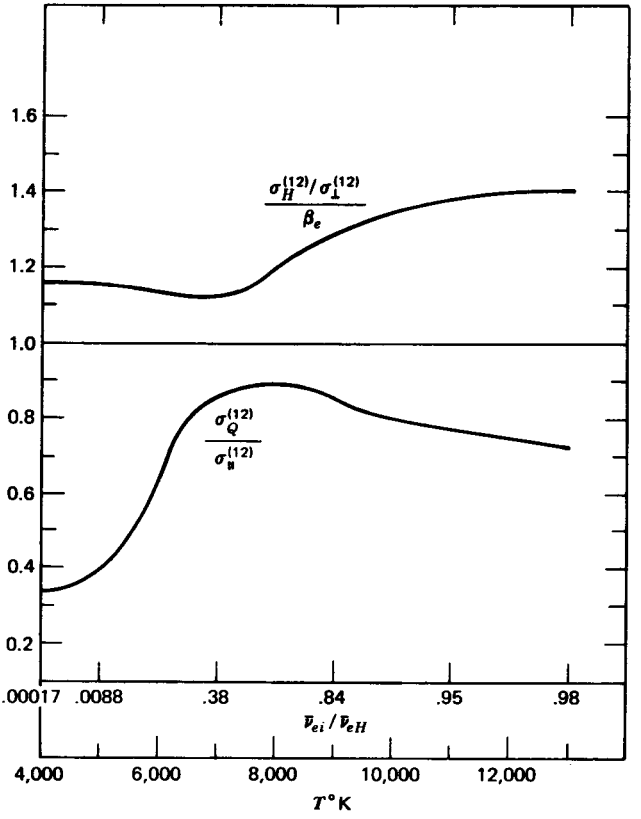



**Figure 11.** Comparison of the exact and mean-free-path expressions for $\sigma_H/\sigma_\perp$ and the quasiscalar conductivity $\sigma_Q = (\sigma_\perp^2 + \sigma_H^2)/\sigma_\perp$ for atmospheric-pressure argon at $\beta_e = \omega_e/\bar{\nu}_{eH} = 5$. According to the mean-free-path representation (7.8), the ratios given by both curves should be everywhere equal to unity.

and accelerators. On the other hand, if the effects of thermal diffusion are neglected, the rigorous expression (4.11) for the current density is

$$\mathbf{J}_e = \sigma_\parallel \mathscr{E}_\parallel + \sigma_\perp \mathscr{E}_\perp + \sigma_H \mathbf{b} \times \mathscr{E}. \tag{7.9}$$

For comparison with equation (7.8), equation (7.9) can be rearranged to yield (see Exercise 7.2)

$$\mathbf{J}_e = \sigma_\parallel \mathscr{E}_\parallel + \left(\frac{\sigma_\perp^2 + \sigma_H^2}{\sigma_\perp}\right)\mathscr{E}_\perp - \frac{\sigma_H}{\sigma_\perp}\mathbf{J}_e \times \mathbf{b}. \tag{7.10}$$

Thus the angle between $(\mathbf{J}_e)_\perp$ and $\mathscr{E}_\perp$ is not $\tan^{-1}\beta_e$ but rather $\tan^{-1}(\sigma_H/\sigma_\perp)$ and the conductivity in the direction of $(\mathbf{J}_e)_\perp$ is not $\sigma_\parallel$ but rather $(\sigma_\perp^2 + \sigma_H^2)/\sigma_\perp$, which has been called the "quasiscalar" conductivity $\sigma_Q$. A comparison of $\sigma_H/\sigma_\perp$ with $\beta_e = \omega_e/\bar{\nu}_{eH}$ and $(\sigma_\perp^2 + \sigma_H^2)/\sigma_\perp$ with $\sigma_\parallel$ is shown in Fig. 11 for atmospheric-pressure equilibrium argon at $\beta_e = 5$. It can be seen that the mean-free-path representation for the quantities in question can be significantly in error.

**Exercise 7.1.** Demonstrate that the Frost formulas (7.3), (7.5), and (7.6) for the various transport properties reduce to the exact expressions in the two limits of weakly ionized and fully ionized plasmas.

**Exercise 7.2.** Show that equation (7.10) follows from equation (7.9). [Suggestion: first evaluate $\mathbf{J}_e \times \mathbf{b}$ from equation (7.9).]

**Exercise 7.3.** Plot as a function of $\beta_e$ the values of $(\sigma_H/\sigma_\perp)/\beta_e$ and $\sigma_Q/\sigma_\parallel$ for a fully ionized plasma, with the use of the results given by Fig. 7 of Sec. 4.

## 8. NONELASTIC COLLISIONS

In the foregoing analysis for the electron transport properties, nonelastic collisions have not been considered. Such collisions, however, play an important role in the dynamics of partially ionized gases. For example, as we shall discuss in the following chapter, nonelastic collisions are of primary importance in the electron continuity equation. It is pertinent, therefore, to ask how nonelastic collisions enter the equations of the present analysis and, in particular, to find the additional restrictions which thereby are imposed on the electron transport-property calculations of Sec. 4.

The Cartesian-tensor equations (VII 6.39) and VII (6.40) for $f^0$ and $\mathbf{f}^1$ are, when multiplied by the factors $(1/2)m_e C^2 4\pi C^2\, dC$ and $m_e C(4/3)\pi C^2\, dC$, respectively, energy and momentum equations for electrons of the speed class $dC$. Accordingly, the nonelastic contributions to these equations can be evaluated from a physical basis by adding respectively to the right-hand

sides terms which account for the net rate per unit volume at which electrons of the speed class $dC$ gain energy and momentum as a result of nonelastic collisions, divided by the aforementioned factors. Such a derivation of these terms is in accord with that obtained by accounting initially for nonelastic collisions in the electron Boltzmann equation (Viegas and Kruger, 1968).

On the basis of the foregoing, the nonelastic loss term in equation (VII 6.39) for $f^0$ is given by $-n_e f^0 \nu_{NE}$, where $\nu_{NE}$ is the nonelastic total collision frequency for electrons of the speed (or equivalently energy) in question, summed over all relevant nonelastic processes. Thus $\nu_{NE}$ will be a function of electron energy and, for example, the populations of excited atomic states. The nonelastic gain term is more complicated in that it involves integrals of $f^0$. From considerations of detailed balancing, however, the gain term will be of the same order of magnitude as the loss term if the plasma is not too far from equilibrium. Hence, it is reasonable to estimate the importance of nonelastic collisions in equation (VII 6.39) on the basis of the loss term alone. Accordingly, it follows that the condition

$$\bar{\nu}_{ee} \gg \sum_h m_e \bar{\nu}_{eh}/m_h \text{ [equation (3.8)]}$$

for $f^0$ to be Maxwellian must be augmented by the condition

$$\bar{\nu}_{ee} \gg \bar{\nu}_{NE} \tag{8.1}$$

for electrons of thermal energies.

Even if the inequality (8.1) is satisfied and $f^0$ is Maxwellian, nonelastic collisions may still enter the calculation of $f^0$ through the integrated energy equation for the electron temperature. In the derivation of the energy equation from equation (VII 6.39) the term involving $\nu_{ee}$ integrates identically to zero, and therefore the term involving $\nu_{NE}$ must be retained even when $\bar{\nu}_{ee} \gg \bar{\nu}_{NE}$. The latter term, which accounts for the net rate of nonelastic energy gain by the electrons, must be added to the right-hand side of the electron energy equation (3.16).

If the relevant nonelastic collisions are with atoms in various levels of electronic excitation that communicate energetically only with the free electrons and by the emission and absorption of radiation, we can obtain the net rate $\dot{N}$ per unit volume of nonelastic energy loss from the free electrons by considering an energy balance for the excited atoms. In this context we consider an ion as an atom with an excitation energy equal to the ionization potential $\epsilon_i$. For simplicity, we consider only one species of singly charged ions. Denoting the local net rate per unit volume of radiant energy loss from the plasma by $\dot{R}$ and the net rate per unit volume of increase in energy in excited levels by $\dot{U}$, we have

$$\dot{U} = \dot{N} - \dot{R}.$$

The quantity $\dot{U}$ can be expressed in terms of the net volumetric rate $\dot{n}_k$ of production of excited atoms as

$$\dot{U} = \sum_k \epsilon_k \dot{n}_k,$$

where $\epsilon_k$ is the energy of the $k$th level, measured from the ground level. We now introduce the continuity equations for the excited atoms, viz.

$$\dot{n}_k = \frac{\partial n_k}{\partial t} + \nabla \cdot [n_k(\mathbf{u} + \mathbf{U}_k)]. \tag{8.2}$$

Combining these expressions, we have for the nonelastic energy loss

$$\dot{N} = \dot{R} + \frac{\partial}{\partial t} \sum_k \epsilon_k n_k + \nabla \cdot \sum_k \epsilon_k(\mathbf{u} + \mathbf{U}_k). \tag{8.3}$$

In most cases of interest the dominant term in the foregoing summations is the contribution from the ions. Under these conditions, equation (8.3) becomes

$$\dot{N} \simeq \dot{R} + \epsilon_i \frac{\partial n_i}{\partial t} + \epsilon_i \nabla \cdot [n_i(\mathbf{u} + \mathbf{U}_i)]. \tag{8.4}$$

To obtain the form of the electron energy equation that accounts for nonelastic collisions, the term $-\dot{N}$ must be added to the right-hand side of equation (3.16). With equation (8.4) for $\dot{N}$ it is convenient to use the relations

$$\frac{\partial n_i}{\partial t} + \nabla \cdot [n_i(\mathbf{u} + \mathbf{U}_i)] = \dot{n}_i,$$

$$\frac{\partial n_e}{\partial t} + \nabla \cdot [n_e(\mathbf{u} + \mathbf{U}_e)] = \dot{n}_e,$$

and, by conservation of charge,

$$\dot{n}_e = \dot{n}_i$$

to write $\dot{N}$ in terms of $n_e$ and $-\mathbf{J}_e/e = n_e \mathbf{U}_e$. [These relations are equivalent to more familiar equation $\nabla \cdot \mathbf{j} = -\partial \rho^c/\partial t$, where $\mathbf{j} = e(n_i - n_e)\mathbf{u} + \mathbf{J}_i + \mathbf{J}_e$ is the total current density and $\rho^c = e(n_i - n_e)$ is the charge density.] The energy equation (3.16) is then replaced by

$$\frac{D}{Dt}\left[n_e\left(\frac{3}{2}kT_e + \epsilon_i\right)\right] + n_e\left(\frac{5}{2}kT_e + \epsilon_i\right)\nabla \cdot \mathbf{u} + \nabla \cdot (\mathbf{q}_e - \epsilon_i \mathbf{J}_e/e) - \mathbf{J}_e \cdot \mathbf{E}'$$

$$= n_e \sum_h \frac{3}{2} k(T_h - T_e) 2 \frac{m_e}{m_h} \bar{\nu}_{eh} - \dot{R}. \tag{8.5}$$

We now turn our attention to the possible nonelastic effects on the calculation of electron transport properties, when $f^0 = f_M$. In equations (VII 6.40) and (4.1) for $\mathbf{f}^1$, the nonelastic loss term is given by $-n_e \mathbf{f}^1 \nu_{NE}$. Insofar as the nonelastic cross sections can be regarded as isotropic functions of scattering direction, the nonelastic gain term will be zero. The analysis for the transport properties would then remain the same as before, with the elastic momentum-transfer collision frequency $\nu_{eH}$ replaced by the sum $\nu_{eH} + \nu_{NE}$. Whenever the inequality (8.1) is satisfied, however, $\nu_{NE}$ will make a negligible contribution to this sum since, as a result of the ion term, $\nu_{eH}$ is at least as great as $\nu_{ee}$. This conclusion applies even if the nonelastic cross sections are not isotropic, if again it is assumed that the nonelastic gain term is of no greater magnitude than the loss term. Thus, if the inequality (8.1) is satisfied, although important nonelastic effects may enter the integrated electron continuity and energy equations, the electron transport-property calculations of Sec. 4 apply without modification.

In the following chapter we shall discuss the effects of nonelastic collisions in greater detail, with an emphasis there on departures from the Saha equation and the rates of ionization and recombination.

**Exercise 8.1.** Obtain the generalization of the electron energy equation (8.5) that accounts in the nonelastic loss term (8.4) for more than one species of singly charged ions, each with a different ionization potential.

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